What is ETL?

ETL is defined as a process that extracts the data from different RDBMS source systems, then transforms the data (like applying calculations, concatenations, etc.) and finally loads the data into the Data Warehouse system. ETL full-form is Extract, Transform and Load.

It's tempting to think a creating a Data warehouse is simply extracting data from multiple sources and loading into database of a Data warehouse. This is far from the truth and requires a complex ETL process. The ETL process requires active inputs from various stakeholders including developers, analysts, testers, top executives and is technically challenging.

In order to maintain its value as a tool for decision-makers, Data warehouse system needs to change with business changes. ETL is a recurring activity (daily, weekly, monthly) of a Data warehouse system and needs to be agile, automated, and well documented.

Why do you need ETL?

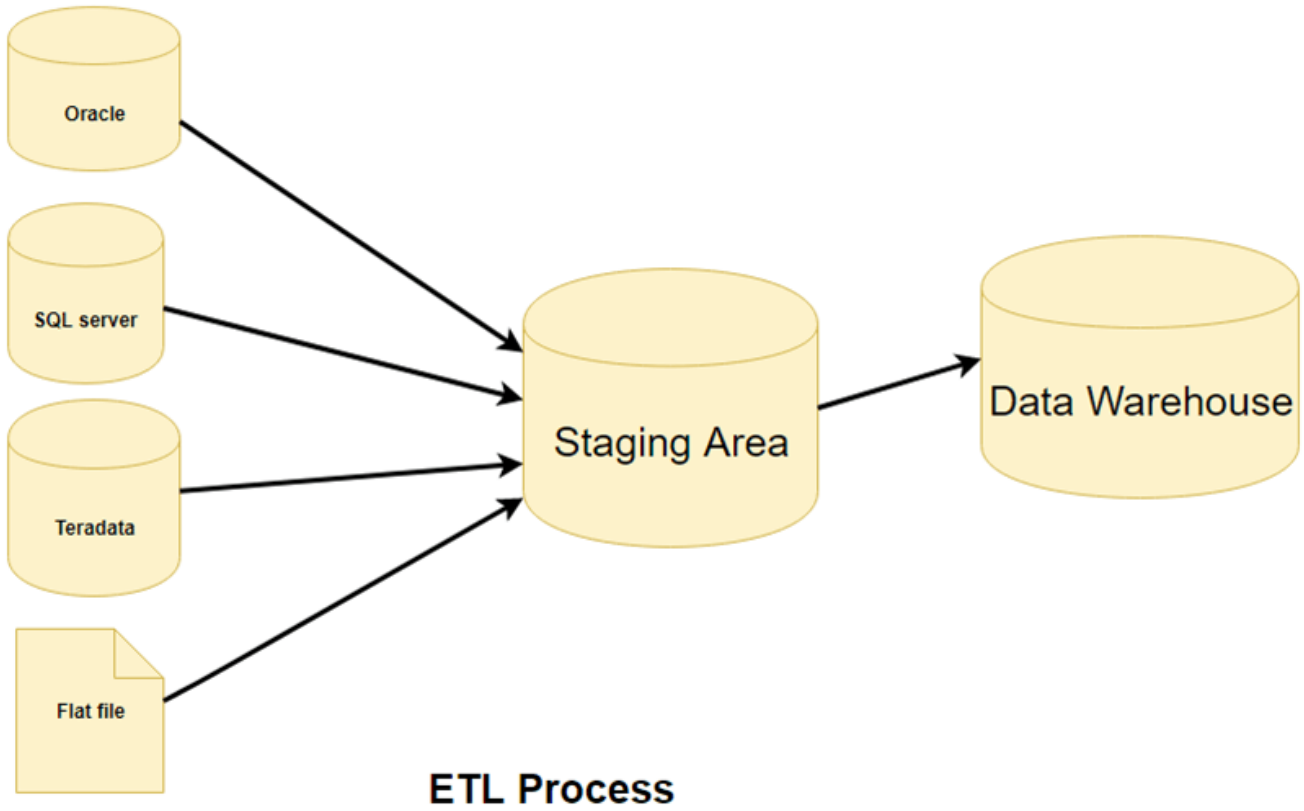
There are many reasons for adopting ETL in the organization:

- It helps companies to analyze their business data for taking critical business decisions.
- Transactional databases cannot answer complex business questions that can be answered by ETL.
- A Data Warehouse provides a common data repository
- ETL provides a method of moving the data from various sources into a data warehouse.

- As data sources change, the Data Warehouse will automatically update.
- Well-designed and documented ETL system is almost essential to the success of a Data Warehouse project.
- Allow verification of data transformation, aggregation and calculations rules.
- ETL process allows sample data comparison between the source and the target system.
- ETL process can perform complex transformations and requires the extra area to store the data.
- ETL helps to Migrate data into a Data Warehouse. Convert to the various formats and types to adhere to one consistent system.
- ETL is a predefined process for accessing and manipulating source data into the target database.
- ETL offers deep historical context for the business.
- It helps to improve productivity because it codifies and reuses without a need for technical skills.

ETL Process in Data Warehouses

ETL is a 3-step process



Step 1) Extraction

In this step, data is extracted from the source system into the staging area. Transformations if any are done in staging area so that performance of source system in not degraded. Also, if corrupted data is copied directly from the source into Data warehouse database, rollback will be a challenge. Staging area gives an opportunity to validate extracted data before it moves into the Data warehouse.

Data warehouse needs to integrate systems that have different

DBMS, Hardware, Operating Systems and Communication Protocols. Sources could include legacy applications like

Mainframes, customized applications, Point of contact devices like ATM, Call switches, text files, spreadsheets, ERP, data from vendors, partners amongst others.

Hence one needs a logical data map before data is extracted and loaded physically. This data map describes the relationship between sources and target data.

Three Data Extraction methods:

1. Full Extraction
2. Partial Extraction- without update notification.
3. Partial Extraction- with update notification

Irrespective of the method used, extraction should not affect performance and response time of the source systems. These source systems are live production databases. Any slow down or locking could effect company's bottom line.

Some validations are done during Extraction:

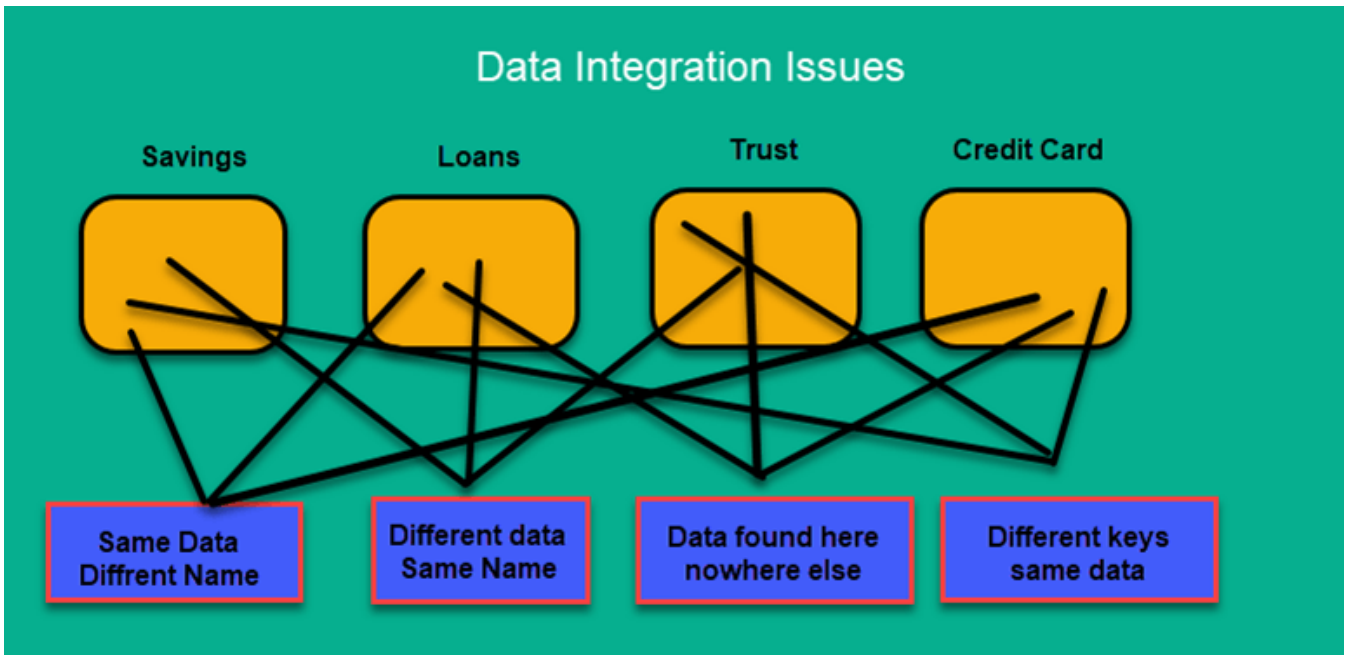
- Reconcile records with the source data
- Make sure that no spam/unwanted data loaded
- Data type check
- Remove all types of duplicate/fragmented data
- Check whether all the keys are in place or not

Step 2) Transformation

Data extracted from source server is raw and not usable in its original form. Therefore it needs to be cleansed, mapped and transformed. In fact, this is the key step where ETL process adds value and changes data such that insightful BI reports can be generated.

In this step, you apply a set of functions on extracted data. Data that does not require any transformation is called as **direct move** or **pass through data**.

In transformation step, you can perform customized operations on data. For instance, if the user wants sum-of-sales revenue which is not in the database. Or if the first name and the last name in a table is in different columns. It is possible to concatenate them before loading.



Following are Data Integrity Problems:

1. Different spelling of the same person like Jon, John, etc.
2. There are multiple ways to denote company name like Google, Google Inc.
3. Use of different names like Cleaveland, Cleveland.
4. There may be a case that different account numbers are generated by various applications for the same customer.
5. In some data required files remains blank

6. Invalid product collected at POS as manual entry can lead to mistakes.

Validations are done during this stage

- Filtering – Select only certain columns to load
- Using rules and lookup tables for Data standardization
- Character Set Conversion and encoding handling
- Conversion of Units of Measurements like Date Time Conversion, currency conversions, numerical conversions, etc.
- Data threshold validation check. For example, age cannot be more than two digits.
- Data flow validation from the staging area to the intermediate tables.
- Required fields should not be left blank.
- Cleaning (for example, mapping NULL to 0 or Gender Male to "M" and Female to "F" etc.)
- Split a column into multiples and merging multiple columns into a single column.
- Transposing rows and columns,
- Use lookups to merge data
- Using any complex data validation (e.g., if the first two columns in a row are empty then it automatically reject the row from processing)

Step 3) Loading

Loading data into the target datawarehouse database is the last step of the ETL process. In a typical Data warehouse, huge volume of data needs to be loaded in a relatively short period (nights). Hence, load process should be optimized for performance.

In case of load failure, recover mechanisms should be configured to restart from the point of failure without data integrity loss. Data Warehouse admins need to monitor, resume, cancel loads as per prevailing server performance.

Types of Loading:

- **Initial Load** — populating all the Data Warehouse tables
- **Incremental Load** — applying ongoing changes as when needed periodically.
- **Full Refresh** —erasing the contents of one or more tables and reloading with fresh data.

Load verification

- Ensure that the key field data is neither missing nor null.
- Test modeling views based on the target tables.
- Check that combined values and calculated measures.
- Data checks in dimension table as well as history table.
- Check the BI reports on the loaded fact and dimension table.

ETL tools

There are many Data Warehousing tools are available in the market. Here, are some most prominent one:

1) Improvado

Improvado is a "data pipeline" -- pulling data from your marketing platforms (Facebook, Google Analytics, Ad Servers, CRMs and email platforms, etc.) and "piping" it into any data warehouse or visualization tool you choose. Brands and agencies love using Improvado because it saves them hours of manual reporting time and millions in wasted marketing spend.

More Information >>

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2. MarkLogic:

MarkLogic is a data warehousing solution which makes data integration easier and faster using an array of enterprise features. It can query different types of data like documents, relationships, and metadata.

http://developer.marklogic.com/products

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3. Oracle:

Oracle is the industry-leading database. It offers a wide range of choice of Data Warehouse solutions for both on-premises and in the cloud. It helps to optimize customer experiences by increasing operational efficiency.

https://www.oracle.com/index.html

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4. Amazon RedShift:

Amazon Redshift is Datawarehouse tool. It is a simple and cost-effective tool to analyze all types of data using standard SQL and existing BI tools. It also allows running complex queries against petabytes of structured data.

https://aws.amazon.com/redshift/?nc2=h_m1

Here is a complete list of useful Data warehouse Tools.

Best practices ETL process

Never try to cleanse all the data:

Every organization would like to have all the data clean, but most of them are not ready to pay to wait or not ready to wait. To clean it all would simply take too long, so it is better not to try to cleanse all the data.

Never cleanse Anything:

Always plan to clean something because the biggest reason for building the Data Warehouse is to offer cleaner and more reliable data.

Determine the cost of cleansing the data:

Before cleansing all the dirty data, it is important for you to determine the cleansing cost for every dirty data element.

To speed up query processing, have auxiliary views and indexes:

To reduce storage costs, store summarized data into disk tapes. Also, the trade-off between the volume of data to be

stored and its detailed usage is required. Trade-off at the level of granularity of data to decrease the storage costs.

Summary:

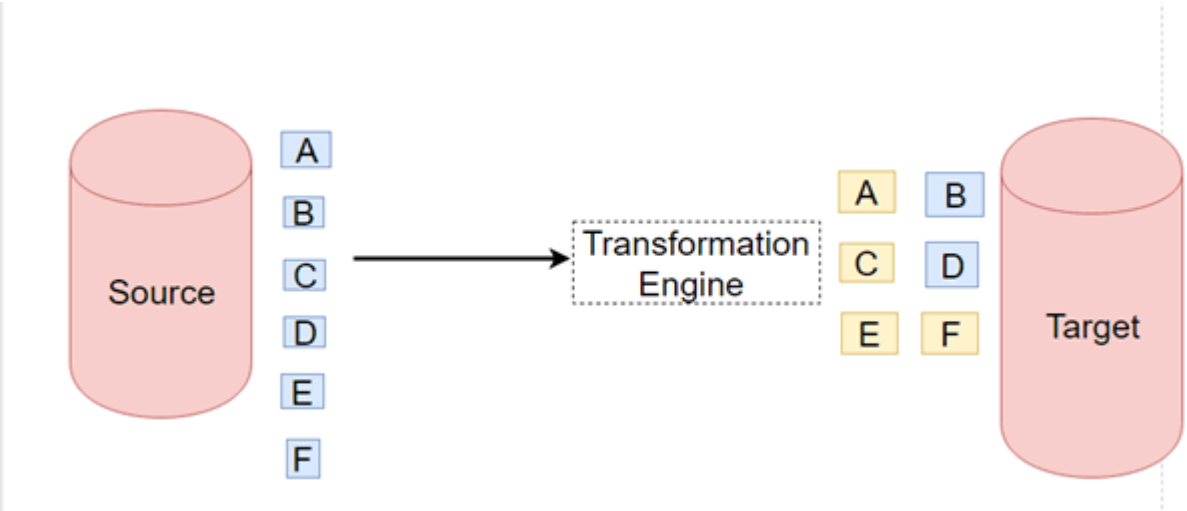
- ETL is an abbreviation of Extract, Transform and Load.
- ETL provides a method of moving the data from various sources into a data warehouse.
- In the first step extraction, data is extracted from the source system into the staging area.
- In the transformation step, the data extracted from source is cleansed and transformed .
- Loading data into the target datawarehouse is the last step of the ETL process

ETL vs ELT:

What is ETL?

ETL is an abbreviation of Extract, Transform and Load. In this process, an ETL tool extracts the data from different RDBMS source systems then transforms the data like applying calculations, concatenations, etc. and then load the data into the Data Warehouse system.

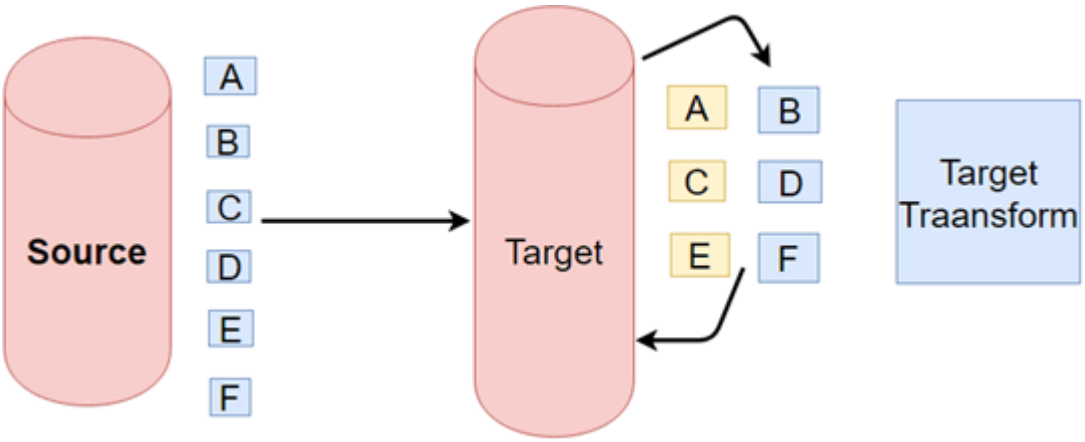
In ETL data is flows from the source to the target. In ETL process transformation engine takes care of any data changes.



What is ELT?

ELT is a different method of looking at the tool approach to data movement. Instead of transforming the data before it's written, ELT lets the target system to do the transformation. The data first copied to the target and then transformed in place.

ELT usually used with no-Sql databases like Hadoop cluster, data appliance or cloud installation.



Difference between ETL vs. ELT

ETL and ELT process are different in following parameters:

Parameters	ETL	ELT
Process	Data is transformed at staging server and then transferred to Datawarehouse DB.	Data remains in the DB of the Datawarehouse.
Code Usage	Used for • Compute-intensive Transformations • Small amount of data	Used for High amounts of data
Transformation	Transformations are done in ETL server/staging area.	Transformations are performed in the target system
Time-Load	Data first loaded into staging and later loaded into target system. Time intensive.	Data loaded into target system only once. Faster.
Time-Transformation	ETL process needs to wait for transformation to complete. As data size grows, transformation time increases.	In ELT process, speed is never dependant on the size of the data.
Time-Maintenance	It needs highs maintenance as you need to select data to load and transform.	Low maintenance as data is always available.
Implementation Complexity	At an early stage, easier to implement.	To implement ELT process organization

Parameters	ETL	ELT
		should have deep knowledge of tools and expert skills.
Support for Data warehouse	ETL model used for on-premises, relational and structured data.	Used in scalable cloud infrastructure which supports structured, unstructured data sources.
Data Lake Support	Does not support.	Allows use of Data lake with unstructured data.
Complexity	The ETL process loads only the important data, as identified at design time.	This process involves development from the output-backward and loading only relevant data.
Cost	High costs for small and medium businesses.	Low entry costs using online Software as a Service Platforms.
Lookups	In the ETL process, both facts and dimensions need to be available in staging area.	All data will be available because Extract and load occur in one single action.
Aggregations	Complexity increase with the additional amount of data in the dataset.	Power of the target platform can process significant amount of data quickly.
Calculations	Overwrites existing column or Need to append the dataset and push to the target platform.	Easily add the calculated column to the existing table.

Parameters	ETL	ELT
Maturity	The process is used for over two decades. It is well documented and best practices easily available.	Relatively new concept and complex to implement.
Hardware	Most tools have unique hardware requirements that are expensive.	Being Saas hardware cost is not an issue.
Support for Unstructured Data	Mostly supports relational data	Support for unstructured data readily available.

KEY DIFFERENCE

- ETL stands for Extract, Transform and Load while ELT stands for Extract, Load, Transform.
- ETL loads data first into the staging server and then into the target system whereas ELT loads data directly into the target system.
- ETL model is used for on-premises, relational and structured data while ELT is used for scalable cloud structured and unstructured data sources.
- ETL is mainly used for a small amount of data whereas ELT is used for large amounts of data.
- ETL doesn’t provide data lake supports while ELT provides data lake support.
- ETL is easy to implement whereas ELT requires niche skills to implement and maintain.

With many Database Warehousing tools available in the market, it becomes difficult to select the top tool for your project. Following is a curated list of most popular open source/commercial ETL tools with key features and download links.

1) CData Sync

Easily replicate all of your Cloud/SaaS data to any database or data warehouse in minutes. CData Sync is an easy-to-use data pipeline that helps you consolidate data from any application or data source into your Database or Data Warehouse of choice. Connect the data that powers your business with BI, Analytics, and Machine Learning.

- From: More than 100+ enterprise data sources including popular CRM, ERP, Marketing Automation, Accounting, Collaboration, and more.
- To: Redshift, Snowflake, BigQuery, SQL Server, MySQL, etc.
- Automated intelligent incremental data replication
- Fully customizable ETL/ELT data transformation
- Runs anywhere – On-premise or in the Cloud

More Information >>

2) QuerySurge

QuerySurge is ETL testing solution developed by RTTS. It is built specifically to automate the testing of Data Warehouses & Big Data. It ensures that the data extracted from data sources remains intact in the target systems as well.

Features:

- Improve data quality & data governance
- Accelerate your data delivery cycles
- Helps to automate manual testing effort
- Provide testing across the different platform like Oracle, Teradata, IBM,

Amazon, Cloudera, etc.

- It speeds up testing process up to 1,000 x and also providing up to 100% data coverage
- It integrates an out-of-the-box DevOps solution for most Build, ETL & QA management software
- Deliver shareable, automated email reports and data health dashboards

More Information >>

3) Xplenty

Xplenty is a cloud-based ETL solution providing simple visualized data pipelines for automated data flows across a wide range of sources and destinations. The company's powerful on-platform transformation tools allow its customers to clean, normalize and transform their data while also adhering to compliance best practices.

Xplenty

Features

- Centralize and prepare data for BI
- Transfer and transform data between internal databases or data warehouses
- Send additional third-party data to Heroku Postgres (and then to Salesforce via Heroku Connect) or directly to Salesforce.
- Rest API connector to pull in data from any Rest API.

More Information >>

4) Vertica

Vertica's SQL analytical database is trusted by the world's leading data-driven companies, including AT&T, Bank of America, Cerner, Guess, Intuit, Phillips

Healthcare, Uber and hundreds more to deliver speed, scale and reliability on mission-critical analytics.

Features:

- High-performance, massively parallel processing SQL query engine with advanced analytics
- In-database machine learning so you can unlock the true potential of your data with no limits and no compromises.
- Whether your organization is shifting all of its data to the cloud or taking a hybrid approach, Vertica provides the most performant analytical database with complete flexibility for on-premise, big data cloud and hybrid deployments.

More Information >>

5) Skyvia

Skyvia is a cloud ETL solution that helps you to quickly get your data from various cloud applications and on-premises database to cloud data warehousing services, such as Amazon Redshift, Google BigQuery, and Azure SQL Data Warehouse.

SKYVIA
STREAM YOUR DATA VIA THE SKY

- Wizard-based, no-coding integration configuration that does not require much technical knowledge
- Configure and automate data loading from cloud apps to data warehouses in just a couple of minutes
- Automatic creation of target tables
- Incremental updates to keep your data warehouse up-to-date
- Powerful data filtering
- Wide support for different cloud apps and databases
- Ability to load data in reverse direction – from data warehouse to cloud

apps and databases

- Requires only a web browser – no need to install anything or have IT infrastructure

More Information >>

6) CloverDX

CloverDX is a data integration platform made for those who demand full, fine control over what they do, who need to solve complex problems in intensive environments, and who prefer to buy best-of-breed tools instead of developing their own.

Features:

- Automate & orchestrate transformations and processes
- Host in cloud or on-premise, scale across cores or cluster nodes
- Code where needed
- Collaborate between devs & less expensive teams
- Co-exist nicely with existing complex IT environment
- Build extensible frameworks to save money and share with colleagues
- Enjoy enterprise grade personal support from CloverDX

More Information >>

7) Panoply

Panoply is a smart data warehouse that automates all three key aspects of the data analytics stack: data collection & transformation (ETL), database storage management, and query performance optimization

- Over 100 pre-built automated data source integrations
- Auto-scales by automatically allocating storage and compute requirements in real-time
- Machine learning to automates joins and builds schemas
- Automated query performance optimization with automated query materialization
- Codeless management UI empowers non-technical users
- Built on AWS architecture and SOC 2 certified and practices HIPAA guidelines
- Connects to any BI visualization tool, e.g. Chartio, Looker, Tableau, PowerBI

More Information >>

8) Oracle:

Oracle data warehouse software is a collection of data which is treated as a unit. The purpose of this database is to store and retrieve related information. It helps the server to reliably manage huge amounts of data so that multiple users can access the same data.

Features:

- Distributes data in the same way across disks to offer uniform performance
- Works for single-instance and real application clusters
- Offers real application testing
- Common architecture between any Private Cloud and Oracle's public cloud

- Hi-Speed Connection to move large data
- Works seamlessly with UNIX/Linux and Windows platforms
- It provides support for virtualization
- Allows connecting to the remote database, table, or view

Download Link: https://www.oracle.com/downloads/index.html

9) Amazon RedShift:

Amazon Redshift is an easy to manage, simple, and cost-effective data warehouse tool. It can analyze almost every type of data using standard SQL.

Features:

- No Up-Front Costs for its installation
- It allows automating most of the common administrative tasks to monitor, manage, and scale your data warehouse
- Possible to change the number or type of nodes
- Helps to enhance the reliability of the data warehouse cluster
- Every data center is fully equipped with climate control
- Continuously monitors the health of the cluster. It automatically re-replicates data from failed drives and replaces nodes when needed

Download Link: https://aws.amazon.com/redshift/

10) Domo:

Domo is a cloud-based Data warehouse management tool that easily integrates various types of data sources, including spreadsheets, databases, social media and almost all cloud-based or on-premise Data warehouse solutions.

Features:

- Help you to build your dream dashboard
- Stay connected anywhere you go
- Integrates all existing business data
- Helps you to get true insights into your business data
- Connects all of your existing business data
- Easy Communication & messaging platform
- It provides support for ad-hoc queries using SQL
- It can handle most concurrent users for running complex and multiple queries

Download Link: https://www.domo.com/product

11) Teradata Corporation:

The Teradata Database is the only commercially available shared-nothing or Massively Parallel Processing (MPP) data warehousing tool. It is one of the best data warehousing tool for viewing and managing large amounts of data.

Features:

- Simple and Cost Effective solutions
- The tool is best suitable option for organization of any size
- Quick and most insightful analytics
- Get the same Database on multiple deployment options
- It allows multiple concurrent users to ask complex questions related to data

- It is entirely built on a parallel architecture
- Offers High performance, diverse queries, and sophisticated workload management

Download Link: https://downloads.teradata.com/

12) SAP:

SAP is an integrated data management platform, to maps all business processes of an organization. It is an enterprise level application suite for open client/server systems. It has set new standards for providing the best business information management solutions.

Features:

- It provides highly flexible and most transparent business solutions
- The application developed using SAP can integrate with any system
- It follows modular concept for the easy setup and space utilization
- You can create a Database system that combines analytics and transactions. These next next-generation databases can be deployed on any device
- Provide support for On-premise or cloud deployment
- Simplified data warehouse architecture
- Integration with SAP and non-SAP applications

Download Link: https://support.sap.com/en/my-support/software-downloads.html

13) SAS:

SAS is a leading Datawarehousing tool that allows accessing data across multiple sources. It can perform sophisticated analyses and deliver information across the organization.

Features:

- Activities managed from central locations. Hence, user can access applications remotely via the Internet
- Application delivery typically closer to a one-to-many model instead of one-to-one model
- Centralized feature updating, allows the users to download patches and upgrades.
- Allows viewing raw data files in external databases
- Manage data using tools for data entry, formatting, and conversion
- Display data using reports and statistical graphics

Download Link: https://www.sas.com/en_in/home.html

14) IBM – DataStage:

IBM data Stage is a business intelligence tool for integrating trusted data across various enterprise systems. It leverages a high-performance parallel framework either in the cloud or on-premise. This data warehousing tool supports extended metadata management and universal business connectivity.

Features:

- Support for Big Data and Hadoop
- Additional storage or services can be accessed without need to install new software and hardware
- Real time data integration
- Provide trusted ETL data anytime, anywhere
- Solve complex big data challenges
- Optimize hardware utilization and prioritize mission-critical tasks
- Deploy on-premises or in the cloud

Download
Link: http://www-01.ibm.com/support/docview.wss?uid=swg24037518

15) Informatica:

Informatica PowerCenter is Data Integration tool developed by Informatica Corporation. The tool offers the capability to connect & fetch data from different sources.

Features:

- It has a centralized error logging system which facilitates logging errors and rejecting data into relational tables
- Build in Intelligence to improve performance
- Limit the Session Log
- **Ability to Scale up Data Integration**
- Foundation for Data Architecture Modernization
- Better designs with enforced best practices on code development
- Code integration with external Software Configuration tools
- Synchronization amongst geographically distributed team members

Download link: https://informatica.com/

16) MS SSIS:

SQL Server Integration Services is a Data warehousing tool that used to perform ETL operations; i.e. extract, transform and load data. SQL Server Integration also includes a rich set of built-in tasks.

Features:

- Tightly integrated with Microsoft Visual Studio and SQL Server
- Easier to maintain and package configuration
- Allows removing network as a bottleneck for insertion of data
- Data can be loaded in parallel and various locations
- It can handle data from different data sources in the same package
- SSIS consumes data which are difficult like FTP, HTTP, MSMQ, and Analysis services, etc.
- Data can be loaded in parallel to many varied destinations

Download link: https://www.microsoft.com/en-us/download/details.aspx?id=39931

17) Talend Open Studio:

Open Studio is an open source data warehousing tool developed by Talend. It is designed to convert, combine and update data in various locations. This tool provides an intuitive set of tools which make dealing with data lot easier. It also allows big data integration, data quality, and master data management.

Features:

- It supports extensive data integration transformations and complex process workflows
- Offers seamless connectivity for more than 900 different databases, files, and applications
- It can manage the design, creation, testing, deployment, etc of integration processes
- Synchronize metadata across database platforms
- Managing and monitoring tools to deploy and supervise the jobs

Download Link: https://www.talend.com/download/

18) The Ab Initio software:

The Ab Initio is a data analysis, batch processing, and GUI based parallel processing data warehousing tool. It is commonly used to extract, transform and load data.

Features:

- Meta data management
- Business and Process Metadata management
- Ability to run, debug Ab Initio jobs and trace execution logs
- Manage and run graphs and control the ETL processes
- Components can execute simultaneously on various branches of a graph

Download Link: https://www.abinitio.com/en/

19) Dundas:

Dundas is an enterprise-ready Business Intelligence platform. It is used for building and viewing interactive dashboards, reports, scorecards and more. It is possible to deploy Dundas BI as the central data portal for the organization or integrate it into an existing website as a custom BI solution.

Features:

- Data warehousing tool for Business Users and IT Professionals
- Easy access through web browser
- Allows to use sample or Excel data
- Server application with full product functionality
- Integrate and access all kind of data sources
- Ad hoc reporting tools
- Customizable data visualizations

- Smart drag and drop tools
- Visualize data through maps
- Predictive and advanced data analytics

Download link: http://www.dundas.com/support/dundas-bi-free-trial

20) Sisense:

Sisense is a business intelligence tool which analyses and visualizes both big and disparate datasets, in real-time. It is an ideal tool for preparing complex data for creating dashboards with a wide variety of visualizations.

Features:

- Unify unrelated data into one centralized place
- Create a single version of truth with seamless data
- Allows to build interactive dashboards with no tech skills
- Query big data at very high speed
- Possible to access dashboards even in the mobile device
- Drag-and-drop user interface
- Eye-grabbing visualization
- Enables to deliver interactive terabyte-scale analytics
- Exports data to Excel, CSV, PDF Images and other formats
- Ad-hoc analysis of high-volume data
- Handles data at scale on a single commodity server
- Identifies critical metrics using filtering and calculations

Download Link: https://www.sisense.com/get/watch-demo/

21) TabLeau:

Tableau Server is an online Data warehousing with 3 versions Desktop, Server, and Online. It is secure, shareable and mobile friendly data warehouse solution.

Features:

- Connect to any data source securely on-premise or in the cloud
- Ideal tool for flexible deployment
- Big data, live or in-memory
- Designed for mobile-first approach
- Securely Sharing and collaborating Data
- Centrally manage metadata and security rules
- Powerful management and monitoring
- Connect to any data anywhere
- Get maximum value from your data with this business analytics platform
- Share and collaborate in the cloud
- Tableau seamlessly integrates with existing security protocols

Download Link: https://public.tableau.com/en-us/s/download

22) MicroStrategy:

MicroStrategy is an enterprise business intelligence application software. This platform supports interactive dashboards, scorecards, highly formatted reports, ad hoc query and automated report distribution.

Features:

- Unmatched speed, performance, and scalability
- Maximize the value of investment made by enterprises
- Eliminating the need to rely on multiple tools
- Support for advanced analytics and big data

- Get insight into complex business processes for strengthening organizational security
- Powerful security and administration feature

Download link: https://www.microstrategy.com/us/get-started

23) Pentaho

Pentaho is a Data Warehousing and Business Analytics Platform. The tool has a simplified and interactive approach which empowers business users to access, discover and merge all types and sizes of data.

Features:

- Enterprise platform to accelerate the data pipeline
- Community Dashboard Editor allows the fast and efficient development and deployment
- Big data integration without a need for coding
- Simplified embedded analytics
- Visualize data with custom dashboards
- Ease of use with the power to integrate all data
- Operational reporting for mongo dB
- Platform to accelerate the data pipeline

Download now: http://www.pentaho.com/testdrive

24) BigQuery:

Google's BigQuery is an enterprise-level data warehousing tool. It reduces the time for storing and querying massive datasets by enabling super-fast SQL queries. It also controls access to both the project and also offering the feature of view or query the data.

Features:

- Offers flexible Data Ingestion
- Read and write data in via Cloud Dataflow, Hadoop, and Spark.
- Automatic Data Transfer Service
- Full control over access to the data stored
- Easy to read and write data in BigQuery via Cloud Dataflow, Spark, and Hadoop
- BigQuery provides cost control mechanisms

Download now: https://cloud.google.com/bigquery/

25) Numetric:

Numetric is the fast and easy BI tool. It offers business intelligence solutions from data centralization and cleaning, analyzing and publishing. It is powerful enough for anyone to use. This data warehousing tool helps to measure and improve productivity.

Features:

- Data benchmarking
- Budgeting & forecasting
- Data chart visualizations
- Data analysis
- Data mapping & dictionary
- Key performance indicators

Download Link: https://www.numetric.com/

26) Solver BI360 Suite:

Solver BI360 is a most comprehensive business intelligence tool. It gives 360º insights into any data, using reporting, data warehousing, and interactive dashboards. BI360 drives effective, data-based productivity.

Features:

- Excel-based reporting with predefined templates
- Currency conversion and inter-company transactions elimination can be automated
- User-friendly budgeting and forecasting feature
- It reduces the amount of time spent for the preparation of reports and planning
- Easy configuration with User-friendly interface
- Automated data loading
- Combine Financial and Operational Data
- Allows to view data in Data Explorer
- Easily add modules and dimensions

- Unlimited Trees on any dimension
- Support for Microsoft SQL Server/SQL Azure

Download link: http://www.solverglobal.com/products/

27) MarkLogic

MarkLogic is a data warehousing solution that makes data integration easier and faster using an array of enterprise features. This tool helps to perform very complex search operations. It can query data including documents, relationships, and metadata.

Features:

- The Optic API can perform joins and aggregates over documents, triples, and rows.
- It allows specifying more complex security rules for all the elements within documents
- Writing, reading, patching, and deleting documents in JSON, XML, text, or binary formats
- Database Replication for Disaster Recovery
- Specify Output Options on the App Server Configuration
- Importing and Exporting Configuration Information

Download Link: https://developer.marklogic.com/products

What is Data Mart?

A data mart is focused on a single functional area of an organization and contains a subset of data stored in a Data Warehouse.

A data mart is a condensed version of Data Warehouse and is designed for use by a specific department, unit or set of users in an organization. E.g., Marketing, Sales, HR or finance. It is often controlled by a single department in an organization.

Data Mart usually draws data from only a few sources compared to a Data warehouse. Data marts are small in size and are more flexible compared to a Datawarehouse.

Type of Data Mart

There are three main types of data marts are:

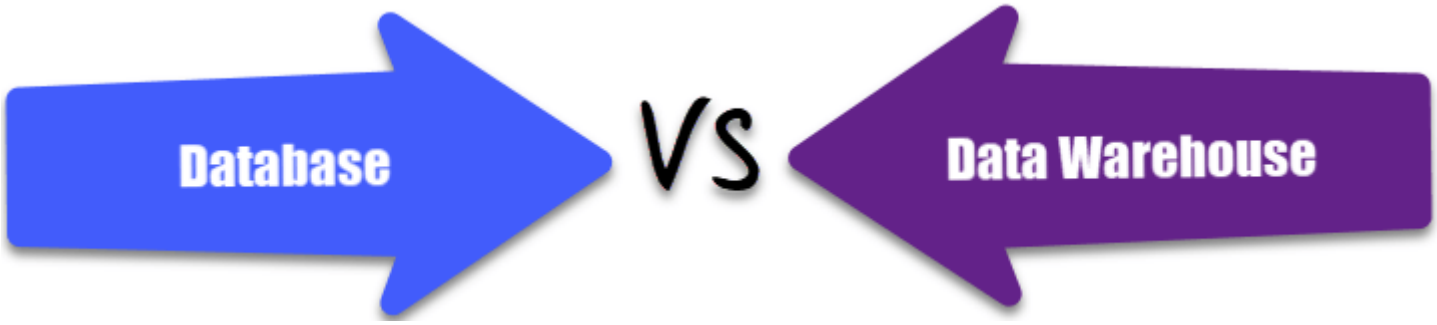
1. **Dependent**: Dependent data marts are created by drawing data directly from operational, external or both sources.
2. **Independent**: Independent data mart is created without the use of a central data warehouse.
3. **Hybrid**: This type of data marts can take data from data warehouses or operational systems.

Examples for Implementing Data Marts

Following are the best practices that you need to follow while in the Data Mart Implementation process:

- The source of a Data Mart should be departmentally structured
- The implementation cycle of a Data Mart should be measured in short periods of time, i.e., in weeks instead of months or years.
- It is important to involve all stakeholders in planning and designing phase as the data mart implementation could be complex.
- Data Mart Hardware/Software, Networking and Implementation costs should be accurately budgeted in your plan
- Even though if the Data mart is created on the same hardware they may need some different software to handle user queries. Additional processing power and disk storage requirements should be evaluated for fast user response
- A data mart may be on a different location from the data warehouse. That's why it is important to ensure that they have enough networking capacity to handle the Data volumes needed to transfer data to the data mart**.**
- Implementation cost should budget the time taken for Datamart loading process. Load time increases with increase in complexity of the transformations.

Difference between Database and Data Warehouse



Parameter	Database	Data Warehouse
Purpose	Is designed to record	Is designed to analyze
Processing Method	The database uses the Online Transactional Processing (OLTP)	Data warehouse uses Online Analytical Processing (OLAP).
Usage	The database helps to perform fundamental operations for your business	Data warehouse allows you to analyze your business.
Tables and	Tables and joins of a	Table and joins are simple

Joins	database are complex as they are normalized.	in a data warehouse because they are denormalized.
Orientation	Is an application-oriented collection of data	It is a subject-oriented collection of data
Storage limit	Generally limited to a single application	Stores data from any number of applications
Availability	Data is available real-time	Data is refreshed from source systems as and when needed
Usage	ER modeling techniques are used for designing.	Data modeling techniques are used for designing.
Technique	Capture data	Analyze data
Data Type	Data stored in the Database is up to date.	Current and Historical Data is stored in Data Warehouse. May not be up to date.

Storage of data	Flat Relational Approach method is used for data storage.	Data Ware House uses dimensional and normalized approach for the data structure. Example: Star and snowflake schema.
Query Type	Simple transaction queries are used.	Complex queries are used for analysis purpose.
Data Summary	Detailed Data is stored in a database.	It stores highly summarized data.

What is Data Warehouse?

A Data Warehouse collects and manages data from varied sources to provide meaningful business insights.

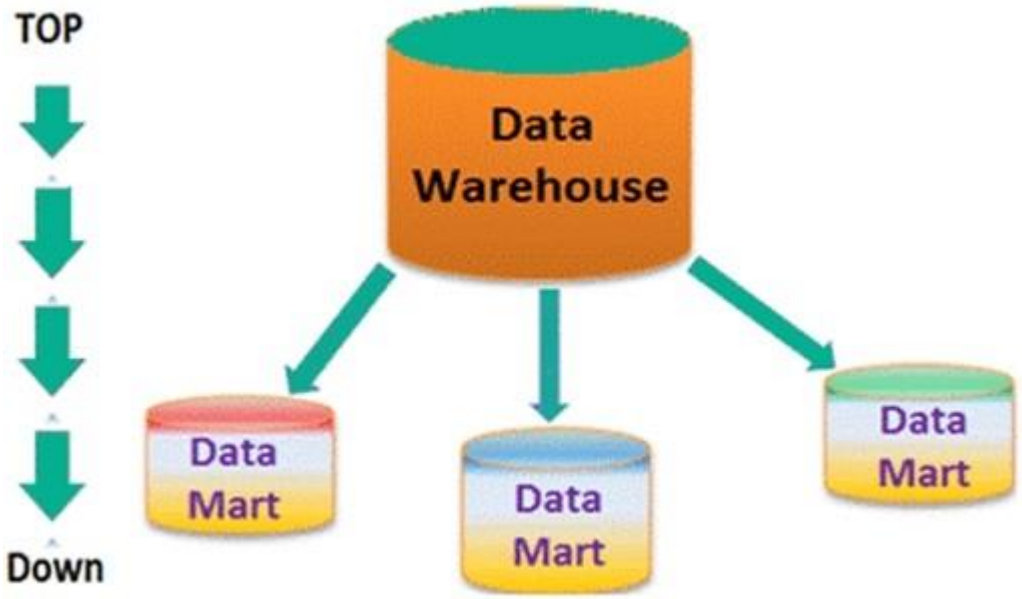
It is a collection of data which is separate from the operational systems and supports the decision making of the company. In Data Warehouse data is stored from a historical perspective.

The data in the warehouse is extracted from multiple functional units. It is checked, cleansed and then integrated with Data warehouse system. Data warehouse used a very fast computer system having large storage capacity. This tool can answer any complex queries relating data.

What is Data Mart?

A data mart is a simple form of a Data Warehouse. It is focused on a single subject. Data Mart draws data from only a few sources. These sources may be central Data warehouse, internal operational systems, or external data sources.

A Data Mart is an index and extraction system. It is an important subset of a data warehouse. It is subject-oriented, and it is designed to meet the needs of a specific group of users. Data marts are fast and easy to use, as they make use of small amounts of data.



Differences between Data Warehouse and Data

Mart

Parameter	Data Warehouse	Data Mart
Definition	A Data Warehouse is a large repository of data collected from different organizations or departments within a corporation.	A data mart is an only subtype c Warehouse. It is designed to me of a certain user group.
Usage	It helps to take a strategic decision.	It helps to take tactical decision business.
Objective	The main objective of Data Warehouse is to provide an integrated environment and coherent picture of the business at a point in time.	A data mart mostly used in a bu division at the department level.
Designing	The designing process of Data Warehouse is quite difficult.	The designing process of Data easy.
	May or may not use in a dimensional model. However, it can feed dimensional models.	It is built focused on a dimensio using a start schema.
Data Handling	Data warehousing includes large area of the corporation which is why it takes a long time to process it.	Data marts are easy to use, des implement as it can only handle amounts of data.
Focus	Data warehousing is broadly focused all the departments. It is possible that it can even represent the entire company.	Data Mart is subject-oriented, a at a department level.
Data type	The data stored inside the Data Warehouse are always detailed when	Data Marts are built for particula groups. Therefore, data short a

	compared with data mart.	
Subject-area	The main objective of Data Warehouse is to provide an integrated environment and coherent picture of the business at a point in time.	Mostly hold only one subject are example, Sales figure.
Data storing	Designed to store enterprise-wide decision data, not just marketing data.	Dimensional modeling and star design employed for optimizing performance of access layer.
Data type	Time variance and non-volatile design are strictly enforced.	Mostly includes consolidation da structures to meet subject area' reporting needs.
Data value	Read-Only from the end-users standpoint.	Transaction data regardless of g directly from the Data Warehou
Scope	Data warehousing is more helpful as it can bring information from any department.	Data mart contains data, of a sp department of a company. Ther maybe separate data marts for finance, marketing, etc. Has lim
Source	In Data Warehouse Data comes from many sources.	In Data Mart data comes from v sources.
Size	The size of the Data Warehouse may range from 100 GB to 1 TB+.	The Size of Data Mart is less th
Implementation time	The implementation process of Data Warehouse can be extended from months to years.	The implementation process of restricted to few months.

KEY DIFFERENCE

- Data Warehouse is a large repository of data collected from different sources whereas Data Mart is only subtype of a data warehouse.
- Data Warehouse is focused on all departments in an organization whereas Data Mart focuses on a specific group.
- Data Warehouse designing process is complicated whereas the Data Mart process is easy to design.
- Data Warehouse takes a long time for data handling whereas Data Mart takes a short time for data handling.
- Data Warehouse size range is 100 GB to 1 TB+ whereas Data Mart size is less than 100 GB.
- Data Warehouse implementation process takes 1 month to 1 year whereas Data Mart takes a few months to complete the implementation process.

What is Data warehouse?

Data warehouse is an information system that contains historical and commutative data from single or multiple sources. It simplifies reporting and analysis process of the organization.

It is also a single version of truth for any company for decision making and forecasting.

What You Will Learn:

What is Data warehouse?

- Characteristics of Data warehouse
  - Subject-Oriented
  - Integrated
  - Time-Variant
  - Non-volatile
- Data Warehouse Architectures
- Datawarehouse Components
  - Data Warehouse Database
  - Sourcing, Acquisition, Clean-up and Transformation Tools (ETL)
  - Metadata
  - Query Tools
  - Data warehouse Bus Architecture
- Data Marts
- Data warehouse Architecture Best Practices

Characteristics of Data warehouse

A data warehouse has following characteristics:

- Subject-Oriented
- Integrated
- Time-variant
- Non-volatile

Subject-Oriented

A data warehouse is subject oriented as it offers information regarding a theme instead of companies' ongoing operations. These subjects can be

sales, marketing, distributions, etc.

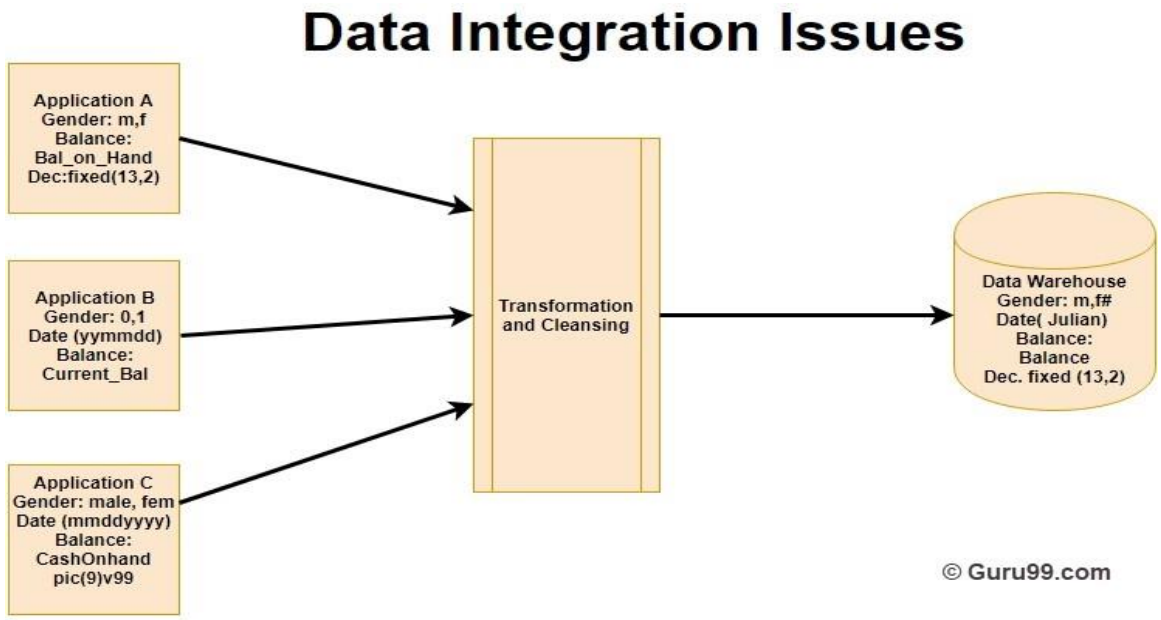
A data warehouse never focuses on the ongoing operations. Instead, it put emphasis on modeling and analysis of data for **decision making**. It also provides a simple and concise view around the specific subject by excluding data which not helpful to support the decision process.

Integrated

In Data Warehouse, integration means the establishment of a common unit of measure for all similar data from the dissimilar database. The data also needs to be stored in the Datawarehouse in common and universally acceptable manner.

A data warehouse is developed by integrating data from varied sources like a mainframe, relational databases, flat files, etc. Moreover, it must keep consistent naming conventions, format, and coding.

This integration helps in effective analysis of data. Consistency in naming conventions, attribute measures, encoding structure etc. have to be ensured. Consider the following example:



In the above example, there are three different application labeled A, B and C. Information stored in these applications are Gender, Date, and Balance. However, each application's data is stored different way.

- In Application A gender field store logical values like M or F
- In Application B gender field is a numerical value,
- In Application C application, gender field stored in the form of a character value.

- Same is the case with Date and balance

However, after transformation and cleaning process all this data is stored in common format in the Data Warehouse.

Time-Variant

The time horizon for data warehouse is quite extensive compared with operational systems. The data collected in a data warehouse is recognized with a particular period and offers information from the historical point of view. It contains an element of time, explicitly or implicitly.

One such place where Datawarehouse data display time variance is in in the structure of the record key. Every primary key contained with the DW should have either implicitly or explicitly an element of time. Like the day, week month, etc.

Another aspect of time variance is that once data is inserted in the warehouse, it can't be updated or changed.

Non-volatile

Data warehouse is also non-volatile means the previous data is not erased when new data is entered in it.

Data is read-only and periodically refreshed. This also helps to analyze historical data and understand what & when happened. It does not require transaction process, recovery and concurrency control mechanisms.

Activities like delete, update, and insert which are performed in an operational application environment are omitted in Data warehouse environment. Only two types of data operations performed in the Data Warehousing are

1. Data loading
2. Data access

Here, are some major differences between Application and Data Warehouse

Operational Application	Data Warehouse
Complex program must be coded to make sure that data upgrade processes maintain high integrity of the final product.	This kind of issues does not happen because data update is not performed.
Data is placed in a normalized form to ensure minimal redundancy.	Data is not stored in normalized form.

Technology needed to support issues of transactions, data recovery, rollback, and resolution as its deadlock is quite complex.	It offers relative simplicity in technology.

Data Warehouse Architectures

There are mainly three types of Datawarehouse Architectures: -

Single-tier architecture

The objective of a single layer is to minimize the amount of data stored. This goal is to remove data redundancy. This architecture is not frequently used in practice.

Two-tier architecture

Two-layer architecture separates physically available sources and data warehouse. This architecture is not expandable and also not supporting a large number of end-users. It also has connectivity problems because of network limitations.

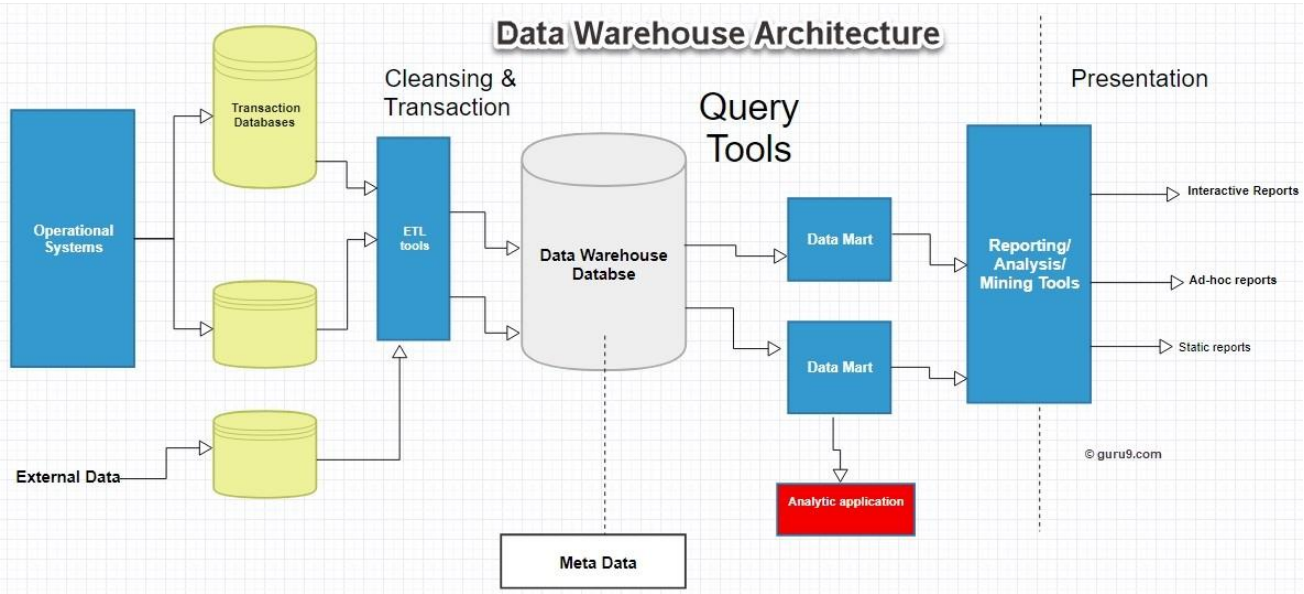
Three-tier architecture

This is the most widely used architecture.

It consists of the Top, Middle and Bottom Tier.

1. **Bottom Tier:** The database of the Datawarehouse servers as the bottom tier. It is usually a relational database system. Data is cleansed, transformed, and loaded into this layer using back-end tools.
2. **Middle Tier:** The middle tier in Data warehouse is an OLAP server which is implemented using either ROLAP or MOLAP model. For a user, this application tier presents an abstracted view of the database. This layer also acts as a mediator between the end-user and the database.
3. **Top-Tier:** The top tier is a front-end client layer. Top tier is the tools and API that you connect and get data out from the data warehouse. It could be Query tools, reporting tools, managed query tools, Analysis tools and Data mining tools.

Datawarehouse Components



The data warehouse is based on an RDBMS server which is a central information repository that is surrounded by some key components to make the entire environment functional, manageable and accessible

There are mainly five components of Data Warehouse:

Data Warehouse Database

The central database is the foundation of the data warehousing environment. This database is implemented on the RDBMS technology. Although, this kind of implementation is constrained by the fact that traditional RDBMS system is optimized for transactional database processing and not for data warehousing. For instance, ad-hoc query, multi-table joins, aggregates are resource intensive and slow down performance.

Hence, alternative approaches to Database are used as listed below-

- In a datawarehouse, relational databases are deployed in parallel to allow for scalability. Parallel relational databases also allow shared memory or shared nothing model on various multiprocessor configurations or massively parallel processors.
- New index structures are used to bypass relational table scan and improve speed.
- Use of multidimensional database (MDDBs) to overcome any limitations which are placed because of the relational data model. Example: Essbase from Oracle.

Sourcing, Acquisition, Clean-up and Transformation Tools (ETL)

The data sourcing, transformation, and migration tools are used for performing all the conversions, summarizations, and all the changes needed to transform data into a unified format in the datawarehouse. They are also called Extract, Transform and Load (ETL) Tools.

Their functionality includes:

- Anonymize data as per regulatory stipulations.
- Eliminating unwanted data in operational databases from loading into Data warehouse.
- Search and replace common names and definitions for data arriving from different sources.
- Calculating summaries and derived data
- In case of missing data, populate them with defaults.
- De-duplicated repeated data arriving from multiple datasources.

These Extract, Transform, and Load tools may generate cron jobs, background jobs, Cobol programs, shell scripts, etc. that regularly update data in datawarehouse. These tools are also helpful to maintain the Metadata.

These ETL Tools have to deal with challenges of Database & Data heterogeneity.

Metadata

The name Meta Data suggests some high- level technological concept. However, it is quite simple. Metadata is data about data which defines the data warehouse. It is used for building, maintaining and managing the data warehouse.

In the Data Warehouse Architecture, meta-data plays an important role as it specifies the source, usage, values, and features of data warehouse data. It also defines how data can be changed and processed. It is closely connected to the data warehouse.

For example, a line in sales database may contain:

```
4030 KJ732 299.90
```

This is a meaningless data until we consult the Meta that tell us it was

- Model number: 4030
- Sales Agent ID: KJ732
- Total sales amount of $299.90

Therefore, Meta Data are essential ingredients in the transformation of data into knowledge.

Metadata helps to answer the following questions

- What tables, attributes, and keys does the Data Warehouse contain?
- Where did the data come from?
- How many times do data get reloaded?
- What transformations were applied with cleansing?

Metadata can be classified into following categories:

1. **Technical Meta Data**: This kind of Metadata contains information about warehouse which is used by Data warehouse designers and administrators.
2. **Business Meta Data:** This kind of Metadata contains detail that gives end-users a way easy to understand information stored in the data warehouse.

Query Tools

One of the primary objects of data warehousing is to provide information to businesses to make strategic decisions. Query tools allow users to interact with the data warehouse system.

These tools fall into four different categories:

1. Query and reporting tools
2. Application Development tools
3. Data mining tools
4. OLAP tools

1. Query and reporting tools:

Query and reporting tools can be further divided into

- Reporting tools
- Managed query tools

Reporting tools: Reporting tools can be further divided into production reporting tools and desktop report writer.

1. Report writers: This kind of reporting tool are tools designed for end-users for their analysis.
2. Production reporting: This kind of tools allows organizations to generate regular operational reports. It also supports high volume batch jobs like printing and calculating. Some popular reporting tools are Brio, Business Objects, Oracle, PowerSoft, SAS Institute.

Managed query tools:

This kind of access tools helps end users to resolve snags in database and SQL and database structure by inserting meta-layer between users and database.

2. Application development tools:

Sometimes built-in graphical and analytical tools do not satisfy the analytical needs of an organization. In such cases, custom reports are developed using Application development tools.

3. Data mining tools:

Data mining is a process of discovering meaningful new correlation, pattens, and trends by mining large amount data. Data mining tools are used to make this process automatic.

4. OLAP tools:

These tools are based on concepts of a multidimensional database. It allows users to analyse the data using elaborate and complex multidimensional views.

Data warehouse Bus Architecture

Data warehouse Bus determines the flow of data in your warehouse. The data flow in a data warehouse can be categorized as Inflow, Upflow, Downflow, Outflow and Meta flow.

While designing a Data Bus, one needs to consider the shared dimensions, facts across data marts.

Data Marts

A data mart is an access layer which is used to get data out to the users. It is presented as an option for large size data warehouse as it takes less time and money to build. However, there is no standard definition of a data mart is differing from person to person.

In a simple word Data mart is a subsidiary of a data warehouse. The data mart is used for partition of data which is created for the specific group of users.

Data marts could be created in the same database as the Datawarehouse or a physically separate Database.

Data warehouse Architecture Best Practices

To design Data Warehouse Architecture, you need to follow below given best practices:

- Use a data model which is optimized for information retrieval which can be the dimensional mode, denormalized or hybrid approach.

- Need to assure that Data is processed quickly and accurately. At the same time, you should take an approach which consolidates data into a single version of the truth.
- Carefully design the data acquisition and cleansing process for Data warehouse.
- Design a MetaData architecture which allows sharing of metadata between components of Data Warehouse
- Consider implementing an ODS model when information retrieval need is near the bottom of the data abstraction pyramid or when there are multiple operational sources required to be accessed.
- One should make sure that the data model is integrated and not just consolidated. In that case, you should consider 3NF data model. It is also ideal for acquiring ETL and Data cleansing tools

Summary:

- Data warehouse is an information system that contains historical and commutative data from single or multiple sources.
- A data warehouse is subject oriented as it offers information regarding subject instead of organization's ongoing operations.
- In Data Warehouse, integration means the establishment of a common unit of measure for all similar data from the different databases
- Data warehouse is also non-volatile means the previous data is not erased when new data is entered in it.
- A Datawarehouse is Time-variant as the data in a DW has high shelf life.
- There are 5 main components of a Datawarehouse. 1) Database 2) ETL Tools 3) Meta Data 4) Query Tools 5) DataMarts
- These are four main categories of query tools 1. Query and reporting, tools 2. Application Development tools, 3. Data mining tools 4. OLAP tools
- The data sourcing, transformation, and migration tools are used for performing all the conversions and summarizations.
- In the Data Warehouse Architecture, meta-data plays an important role as it specifies the source, usage, values, and features of data warehouse data.

Chapter 11

Expert Systems and Applied Artificial Intelligence

11.1 What is Artificial Intelligence?

The field of artificial intelligence (AI) is concerned with methods of developing systems that display aspects of intelligent behaviour. These systems are designed to imitate the human capabilities of thinking and sensing.

Characteristics of AI Systems

Characteristics of AI systems include:

1. Symbolic Processing

In AI applications, computers process symbols rather than numbers or letters. AI applications process strings of characters that represent real-world entities or concepts. Symbols can be arranged in structures such as lists, hierarchies, or networks. These structures show how symbols relate to each other.

2. Nonalgorithmic Processing

Computer programs outside the AI domain are programmed algorithms; that is, fully specified step-by-step procedures that define a solution to the problem. The actions of a knowledge-based AI system depend to a far greater degree on the situation where it is used.

The Field of AI

Artificial intelligence is a science and technology based on disciplines such as computer science, biology, psychology, linguistics, mathematics, and engineering. The goal of AI is to develop computers that can think, see, hear, walk, talk and feel. A major thrust of AI is the development of computer functions normally associated with human intelligence, such as reasoning, learning, and problem solving.

How the AI Field Evolved [Figure 11.2]

1950 Turing Test - a machine performs intelligently if an interrogator using remote terminals cannot distinguish its responses from those of a human.

Result: General problem-solving methods

1960 AI established as research field.

Result: Knowledge-based expert systems

1970 AI commercialization began

Result: Transaction processing and decision support systems using AI.

1980 Artificial neural networks

Result: Resembling the interconnected neuronal structures in the human brain

1990 Intelligent agents

Result: Software that performs assigned tasks on the users behalf

11.2 Capabilities of Expert Systems: General View

The most important applied area of AI is the field of expert systems. An ***expert system*** (ES) is a knowledge-based system that employs knowledge about its application domain and uses an inferencing (reason) procedure to solve problems that would otherwise require human competence or expertise. The power of expert systems stems primarily from the specific knowledge about a narrow domain stored in the expert system's ***knowledge base***.

It is important to stress to students that expert systems are assistants to decision makers and not substitutes for them. Expert systems do not have human capabilities. They use a knowledge base of a particular domain and bring that knowledge to bear on the facts of the particular situation at hand. The knowledge base of an ES also contains ***heuristic knowledge*** **-** rules of thumb used by human experts who work in the domain.

11.3 Applications of Expert Systems

The test outlines some illustrative minicases of expert systems applications. These include areas such as high-risk credit decisions, advertising decision

making, and manufacturing decisions.

Generic Categories of Expert System Applications

Table 11.1 outlines the generic areas of ES applications where ES can be applied. Application areas include classification, diagnosis, monitoring, process control, design, scheduling and planning, and generation of options.

Classification - identify an object based on stated characteristics

Diagnosis Systems - infer malfunction or disease from observable data

Monitoring - compare data from a continually observed system to prescribe behaviour

Process Control - control a physical process based on monitoring

Design - configure a system according to specifications

Scheduling & Planning - develop or modify a plan of action

Generation of Options - generate alternative solutions to a problem

11.4 How Expert Systems Work

The strength of an ES derives from its ***knowledge base*** - an organized collection of facts and heuristics about the system's domain. An ES is built in a process known as ***knowledge engineering***, during which knowledge about the domain is acquired from human experts and other sources by knowledge engineers.

The accumulation of knowledge in knowledge bases, from which conclusions are to be drawn by the inference engine, is the hallmark of an expert system.

Knowledge Representation and the Knowledge Base

The knowledge base of an ES contains both factual and heuristic knowledge. ***Knowledge representation*** is the method used to organize the knowledge in the knowledge base. Knowledge bases must represent notions as actions to be taken under circumstances, causality, time, dependencies, goals, and other higher-level concepts.

Several methods of knowledge representation can be drawn upon. Two of these methods include:

1. Frame-based systems

- are employed for building very powerful ESs. A frame specifies the attributes of a complex object and frames for various object types have specified relationships.

2. Production rules

- are the most common method of knowledge representation used in business. ***Rule-based expert systems*** are expert systems in which the knowledge is represented by production rules.

A production rule, or simply a rule, consists of an IF part (a condition or premise) and a THEN part (an action or conclusion). IF condition THEN action (conclusion).

The ***explanation facility*** explains how the system arrived at the recommendation. Depending on the tool used to implement the expert system, the explanation may be either in a natural language or simply a listing of rule numbers.

Inference Engine [Figure 11.4]

The inference engine:

1. Combines the facts of a specific case with the knowledge contained in the knowledge base to come up with a recommendation. In a rule-based expert system, the inference engine controls the order in which production rules are applied (Afired@) and resolves conflicts if more than one rule is applicable at a given time. This is what Areasoning@ amounts to in rule-based systems.

2. Directs the user interface to query the user for any information it needs for further inferencing.

The facts of the given case are entered into the ***working memory***, which acts as a blackboard, accumulating the knowledge about the case at hand. The inference engine repeatedly applies the rules to the working memory, adding

new information (obtained from the rules conclusions) to it, until a goal state is produced or confirmed.

Figure 11.5 One of several strategies can be employed by an inference engine to reach a conclusion. Inferencing engines for rule-based systems generally work by either forward or backward chaining of rules. Two strategies are:

1. Forward chaining

- is a data-driven strategy. The inferencing process moves from the facts of the case to a goal (conclusion). The strategy is thus driven by the facts available in the working memory and by the premises that can be satisfied. The inference engine attempts to match the condition (IF) part of each rule in the knowledge base with the facts currently available in the working memory. If several rules match, a conflict resolution procedure is invoked; for example, the lowest-numbered rule that adds new information to the working memory is fired. The conclusion of the firing rule is added to the working memory.

Forward-chaining systems are commonly used to solve more open-ended problems of a design or planning nature, such as, for example, establishing the configuration of a complex product.

2. Backward chaining

- the inference engine attempts to match the assumed (hypothesized) conclusion - the goal or subgoal state - with the conclusion (THEN) part of the rule. If such a rule is found, its premise becomes the new subgoal. In an ES with few possible goal states, this is a good strategy to pursue.

If a hypothesized goal state cannot be supported by the premises, the system will attempt to prove another goal state. Thus, possible conclusions are review until a goal state that can be supported by the premises is encountered.

Backward chaining is best suited for applications in which the possible conclusions are limited in number and well defined. Classification or diagnosis type systems, in which each of several possible conclusions can be checked to see if it is supported by the data, are typical applications.

Uncertainty and Fuzzy Logic

Fuzzy logic is a method of reasoning that resembles human reasoning since it allows for approximate values and inferences and incomplete or ambiguous data (fuzzy data). Fuzzy logic is a method of choice for handling uncertainty in some expert systems.

Expert systems with fuzzy-logic capabilities thus allow for more flexible and creative handling of problems. These systems are used, for example, to control manufacturing processes.

11.5 Expert System Technology [Figure 11.6]

There are several levels of ES technologies available. Two important things to keep in mind when selecting ES tools include:

1. The tool selected for the project has to match the capability and sophistication of the projected ES, in particular, the need to integrate it with other subsystems such as databases and other components of a larger information system.

2. The tool also has to match the qualifications of the project team.

Expert systems technologies include:

1. Specific expert systems

- These expert systems actually provide recommendations in a specific task domain.

2. Expert system shells

- are the most common vehicle for the development of specific ESs. A shell is an expert system without a knowledge base. A shell furnishes the ES developer with the inference engine, user interface, and the explanation and knowledge acquisition facilities.

Domain-specific shells are actually incomplete specific expert systems, which require much less effort in order to field an actual system.

3. Expert system development environments

- these systems expand the capabilities of shells in various directions. They run on engineering workstations, minicomputers, or mainframes; offer tight integration with large databases; and support the building of large expert systems.

4. High-level programming languages

Several ES development environments have been rewritten from LISP into a procedural language more commonly found in the commercial environment, such as C or C++. ESs are now rarely developed in a programming language.

11.6 Roles in Expert System Development

Three fundamental roles in building expert systems are:

1. ***Expert*** - Successful ES systems depend on the experience and application of knowledge that the people can bring to it during its development. Large systems generally require multiple experts.

2. ***Knowledge engineer*** - The knowledge engineer has a dual task. This person should be able to elicit knowledge from the expert, gradually gaining an understanding of an area of expertise. Intelligence, tact, empathy, and proficiency in specific techniques of knowledge acquisition are all required of a knowledge engineer. Knowledge-acquisition techniques include conducting interviews with varying degrees of structure, protocol analysis, observation of experts at work, and analysis of cases.

On the other hand, the knowledge engineer must also select a tool appropriate for the project and use it to represent the knowledge with the application of the ***knowledge acquisition facility***.

3. ***User*** - A system developed by an end user with a simple shell, is built rather quickly an inexpensively. Larger systems are built in an organized development effort. A prototype-oriented iterative development strategy is commonly used. ESs lends themselves particularly well to prototyping.

11.7 Development and Maintenance of Expert Systems [Figure 11.7]

Steps in the methodology for the iterative process of ES development and maintenance include:

1. Problem Identification and Feasibility Analysis:

- the problem must be suitable for an expert system to solve it.

- must find an expert for the project

- cost-effectiveness of the system has to be established (feasibility)

2. System Design and ES Technology Identification:

- the system is being designed. The needed degree of integration with other subsystems and databases is established

- concepts that best represent the domain knowledge are worked out

- the best way to represent the knowledge and to perform inferencing should be established with sample cases

3. Development of Prototype:

- knowledge engineer works with the expert to place the initial kernel of knowledge in the knowledge base.

- knowledge needs to be expressed in the language of the specific tool chosen for the project

4. Testing and Refinement of Prototype:

- using sample cases, the prototype is tested, and deficiencies in performance are noted. End users test the prototypes of the ES.

5. Complete and Field the ES:

- the interaction of the ES with all elements of its environment, including users and other information systems, is ensured and tested.

- ES is documented and user training is conducted

6. Maintain the System:

- the system is keep current primarily by updating its knowledge base.

- interfaces with other information systems have to be maintained as well, as those systems evolve.

11-8 Expert Systems in Organizations: Benefits and Limitations

Expert systems offer both tangible and important intangible benefits to owner companies. These benefits should be weighted against the development and exploitation costs of an ES, which are high for large, organizationally important ESs.

Benefits of Expert Systems

An ES is no substitute for a knowledge worker's overall performance of the problem-solving task. But these systems can dramatically reduce the amount of work the individual must do to solve a problem, and they do leave people with the creative and innovative aspects of problem solving.

Some of the possible organizational benefits of expert systems are:

1. An Es can complete its part of the tasks much faster than a human expert.

2. The error rate of successful systems is low, sometimes much lower than the human error rate for the same task.

3. ESs make consistent recommendations

4. ESs are a convenient vehicle for bringing to the point of application difficult-to-use sources of knowledge.

5. ESs can capture the scarce expertise of a uniquely qualified expert.

6. ESs can become a vehicle for building up organizational knowledge, as opposed to the knowledge of individuals in the organization.

7. When use as training vehicles, ESs result in a faster learning curve for novices.

8. The company can operate an ES in environments hazardous for humans.

Limitations of Expert Systems

No technology offers an easy and total solution. Large systems are costly and require significant development time and computer resources. ESs also have their limitations which include:

1. Limitations of the technology

2. Problems with knowledge acquisition

3. Operational domains as the principal area of ES application

4. Maintaining human expertise in organizations

11-9 Overview of Applied Artificial Intelligence

Expert systems are only one area of AI. Other areas include:

1. Natural language processing

2. Robotics

3. Computer vision

4. Computerized speech recognition

5. Machine learning

Natural Language Processing

Being able to talk to computers in conversational human languages and have them Aunderstand@ us in a goal of AI researchers. Natural language processing systems are becoming common. The main application for natural language systems at this time is as a user interface for expert and database systems.

Robotics

AI, engineering, and physiology are the basic disciplines of robotics. This technology produces robot machines with computer intelligence and computer-controlled, human like physical capabilities, robotics applications

Computer Vision

The simulation of human senses is a principal objective of the AI field. The most advanced AI sensory system is compute vision, or visual scene recognition. The task of a vision system is to interpret the picture obtained. These systems are employed in robots or in satellite systems. Simpler vision systems are used for quality control in manufacturing.

Speech Recognition

The ultimate goal of the corresponding AI area is computerized speech recognition, or the understanding of connected speech by an unknown speaker, as opposed to systems that recognize words or short phrases spoken one at a time or systems are trained by a specific speaker before use.

Machine Learning

A system with learning capabilities - machine learning - can automatically change itself in order to perform the same tasks more efficiently and more effectively the next time.

A number of approaches to learning are being investigated. Approaches include:

1. Problem Solving Learning - accumulate Aexperience@ about its rules in terms of their contributions to correct advice. The rules that do not contribute or those that are found to provide doubtful contributions could be automatically discarded or assigned low certainty factors.

2. Case-Based Learning - collecting cases in a knowledge base and solving problems by seeking out a case similar to the one to be solved.

3. Inducive Learning - learning from examples. In this case, a system is able to generate its knowledge, represented as rules.

11-10 Neural Networks

Neural networks are computing systems modelled on the human brain's mesh-like network of interconnected processing elements, called neurons. Of course, neural networks are much simpler than the human brain (estimated to have more than 100 billion neuron brain cells). Like the brain, however, such networks can process many pieces of information simultaneously and can

learn to recognize patterns and programs themselves to solve related problems on their own.

A ***neural network*** is an array of interconnected processing elements, each of which can accept inputs, process them, and produce a single output with the objective of imitating the operation of the human brain. Knowledge is represented in a neural network by the pattern of connections among the processing elements and by adjusting weights of these connections.

The strength of neural networks is in applications that require sophisticated pattern recognition. The greatest weakness of neural networks is that they do not furnish an explanation for the conclusions they make.

In summary, a neural network can be trained to recognize certain patterns and then apply what it learned to new cases where it can discern the patterns.

What is OLAP?

Online Analytical Processing, a category of software tools which provide analysis of data for business decisions. OLAP systems allow users to analyze database information from multiple database systems at one time.

The primary objective is data analysis and not data processing.

What is OLTP?

Online transaction processing shortly known as OLTP supports transaction-oriented applications in a 3-tier architecture. OLTP administers day to day transaction of an organization.

The primary objective is data processing and not data analysis

Example of OLAP

Any Datawarehouse system is an OLAP system. Uses of OLAP are as follows

- A company might compare their mobile phone sales in September with sales in October, then compare those results with another location which may be stored in a sperate database.
- Amazon analyzes purchases by its customers to come up with a personalized homepage with products which likely interest to their customer.

Example of OLTP system

An example of OLTP system is ATM center. Assume that a couple has a joint account with a bank. One day both simultaneously reach different ATM centers at precisely the same time and want to withdraw total amount present in their bank account.

However, the person that completes authentication process first will be able to get money. In this case, OLTP system makes sure that withdrawn amount will be never more than the amount present in the bank. The key to note here is that OLTP systems are optimized for **transactional superiority instead data analysis.**

Other examples of OLTP system are:

- Online banking
- Online airline ticket booking
- Sending a text message

- Order entry
- Add a book to shopping cart

Benefits of using OLAP services

- OLAP creates a single platform for all type of business analytical needs which includes planning, budgeting, forecasting, and analysis.
- The main benefit of OLAP is the consistency of information and calculations.
- Easily apply security restrictions on users and objects to comply with regulations and protect sensitive data.

Benefits of OLTP method

- It administers daily transactions of an organization.
- OLTP widens the customer base of an organization by simplifying individual processes.

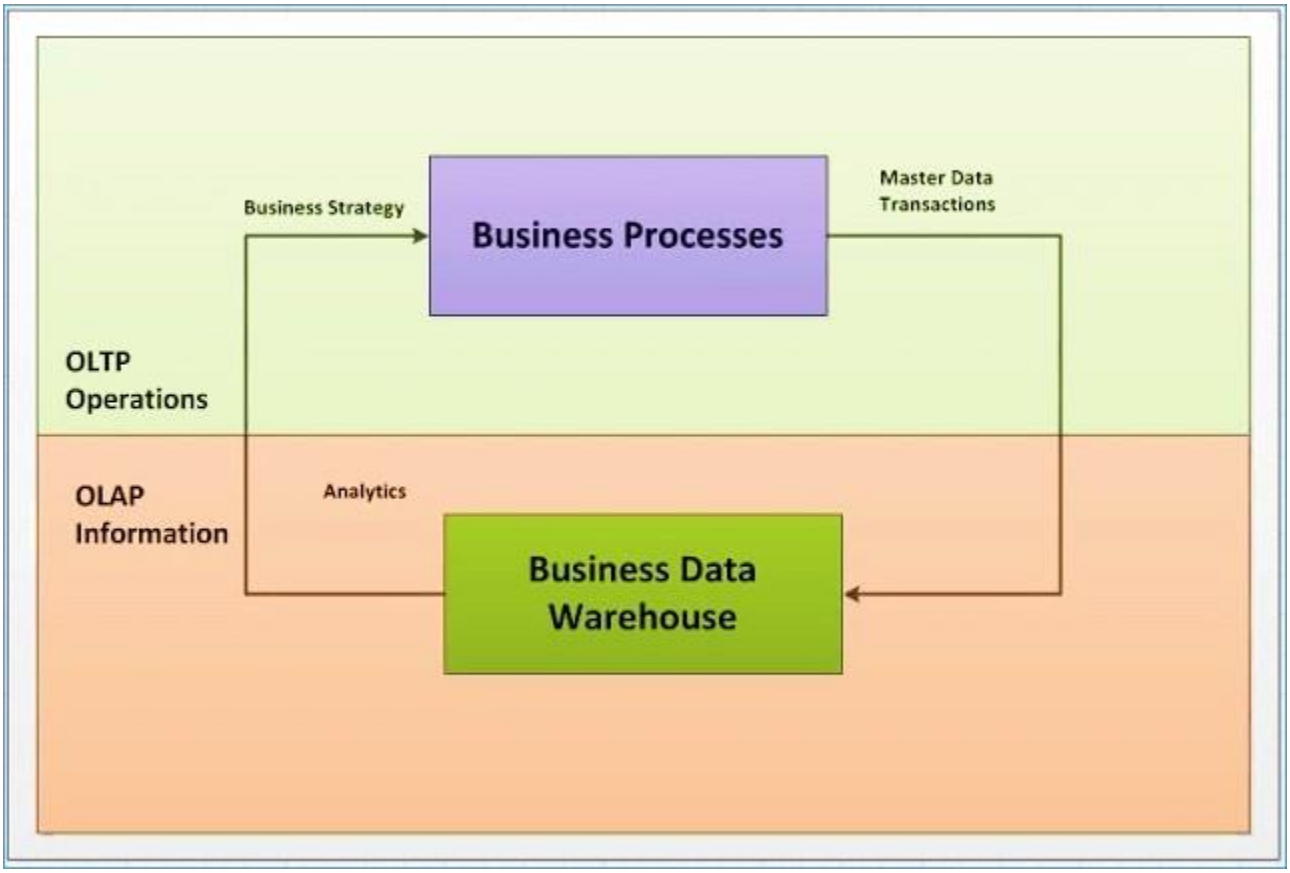
Drawbacks of OLAP service

- Implementation and maintenance are dependent on IT professional because the traditional OLAP tools require a complicated modeling procedure.
- OLAP tools need cooperation between people of various departments to be effective which might always be not possible.

Drawbacks of OLTP method

- If OLTP system faces hardware failures, then online transactions get severely affected.
- OLTP systems allow multiple users to access and change the same data at the same time which many times created unprecedented situation.

Difference between OLTP and OLAP



OLTP vs OLAP

Parameters	OLTP	OLAP
Process	It is an online transactional system. It manages database modification.	OLAP is an online analysis and dat process.
Characteristic	It is characterized by large numbers of short online transactions.	It is characterized by a large volum
Functionality	OLTP is an online database modifying system.	OLAP is an online database query management system.
Method	OLTP uses traditional DBMS.	OLAP uses the data warehouse.
Query	Insert, Update, and Delete	Mostly select operations

	information from the database.	
Table	Tables in OLTP database are normalized.	Tables in OLAP database are **not** n
Source	OLTP and its transactions are the sources of data.	Different OLTP databases become of data for OLAP.
Data Integrity	OLTP database must maintain data integrity constraint.	OLAP database does not get frequ modified. Hence, data integrity is n
Response time	It's response time is in millisecond.	Response time in seconds to minut
Data quality	The data in the OLTP database is always detailed and organized.	The data in OLAP process might n organized.
Usefulness	It helps to control and run fundamental business tasks.	It helps with planning, problem-solv decision support.
Operation	Allow read/write operations.	Only read and rarely write.
Audience	It is a market orientated process.	It is a customer orientated process
Query Type	Queries in this process are standardized and simple.	Complex queries involving aggrega
Back-up	Complete backup of the data combined with incremental backups.	OLAP only need a backup from tim Backup is not important compared
Design	DB design is application oriented. Example: Database design	DB design is subject oriented. Exa Database design changes with sub

	changes with industry like Retail, Airline, Banking, etc.	sales, marketing, purchasing, etc.
User type	It is used by Data critical users like clerk, DBA & Data Base professionals.	Used by Data knowledge users like managers, and CEO.
Purpose	Designed for real time business operations.	Designed for analysis of business by category and attributes.
Performance metric	Transaction throughput is the performance metric	Query throughput is the performan
Number of users	This kind of Database users allows thousands of users.	This kind of Database allows only h users.
Productivity	It helps to Increase user's self-service and productivity	Help to Increase productivity of the analysts.
Challenge	Data Warehouses historically have been a development project which may prove costly to build.	An OLAP cube is not an open SQL warehouse. Therefore, technical kn and experience is essential to man OLAP server.
Process	It provides fast result for daily used data.	It ensures that response to the que quicker consistently.
Characteristic	It is easy to create and maintain.	It lets the user create a view with th spreadsheet.
Style	OLTP is designed to have fast response time, low data redundancy and is normalized.	A data warehouse is created uniqu can integrate different data sources building a consolidated database

KEY DIFFERENCE:

- Online Analytical Processing (OLAP) is a category of software tools that analyze data stored in a database whereas Online transaction processing (OLTP) supports transaction-oriented applications in a 3-tier architecture.
- OLAP creates a single platform for all type of business analysis needs which includes planning, budgeting, forecasting, and analysis while OLTP is useful to administer day to day transactions of an organization.
- OLAP is characterized by a large volume of data while OLTP is characterized by large numbers of short online transactions.
- In OLAP, data warehouse is created uniquely so that it can integrate different data sources for building a consolidated database whereas OLTP uses traditional DBMS.

What is Data Warehousing?

A **Data Warehousing** (DW) is process for collecting and managing data from varied sources to provide meaningful business insights. A Data warehouse is typically used to connect and analyze business data from heterogeneous sources. The data warehouse is the core of the BI system which is built for data analysis and reporting.

It is a blend of technologies and components which aids the strategic use of data. It is electronic storage of a large amount of information by a business which is designed for query and analysis instead of transaction processing. It is a process of transforming data into information and making it available to users in a timely manner to make a difference.

The decision support database (Data Warehouse) is maintained separately from the organization's operational database. However, the data warehouse is not a product but an environment. It is an architectural construct of an information system which provides users with current and historical decision support information which is difficult to access or present in the traditional operational data store.

You many know that a 3NF-designed database for an inventory system many have tables related to each other. For example, a report on current inventory information can include more than 12 joined conditions. This can quickly slow down the response time of the query and report. A data warehouse provides a new design which can help to reduce the response time and helps to enhance the performance of queries for reports and analytics.

Data warehouse system is also known by the following name:

- Decision Support System (DSS)
- Executive Information System
- Management Information System
- Business Intelligence Solution
- Analytic Application
- Data Warehouse

History of Datawarehouse

The Datawarehouse benefits users to understand and enhance their organization's performance. The need to warehouse data evolved as computer systems became more complex and needed to handle increasing amounts of Information. However, Data Warehousing is a not a new thing.

Here are some key events in evolution of Data Warehouse-

- 1960- Dartmouth and General Mills in a joint research project, develop the terms dimensions and facts.
- 1970- A Nielsen and IRI introduces dimensional data marts for retail sales.
- 1983- Tera Data Corporation introduces a database management system which is specifically designed for decision support
- Data warehousing started in the late 1980s when IBM worker Paul Murphy and Barry Devlin developed the Business Data Warehouse.
- However, the real concept was given by Inmon Bill. He was considered as a father of data warehouse. He had written about a variety of topics

for building, usage, and maintenance of the warehouse & the Corporate Information Factory.

How Datawarehouse works?

A Data Warehouse works as a central repository where information arrives from one or more data sources. Data flows into a data warehouse from the transactional system and other relational databases.

Data may be:

1. Structured
2. Semi-structured
3. Unstructured data

The data is processed, transformed, and ingested so that users can access the processed data in the Data Warehouse through Business Intelligence tools, SQL clients, and spreadsheets. A data warehouse merges information coming from different sources into one comprehensive database.

By merging all of this information in one place, an organization can analyze its customers more holistically. This helps to ensure that it has considered all the information available. Data warehousing makes data mining possible. Data mining is looking for patterns in the data that may lead to higher sales and profits.

Types of Data Warehouse

Three main types of Data Warehouses are:

1. Enterprise Data Warehouse:

Enterprise Data Warehouse is a centralized warehouse. It provides decision support service across the enterprise. It offers a unified approach for organizing and representing data. It also provide the ability to classify data according to the subject and give access according to those divisions.

2. Operational Data Store:

Operational Data Store, which is also called ODS, are nothing but data store required when neither Data warehouse nor OLTP systems support organizations reporting needs. In ODS, Data warehouse is refreshed in real time. Hence, it is widely preferred for routine activities like storing records of the Employees.

3. Data Mart:

A data mart is a subset of the data warehouse. It specially designed for a particular line of business, such as sales, finance, sales or finance. In an

independent data mart, data can collect directly from sources.

General stages of Data Warehouse

Earlier, organizations started relatively simple use of data warehousing. However, over time, more sophisticated use of data warehousing begun.

The following are general stages of use of the data warehouse:

Offline Operational Database:

In this stage, data is just copied from an operational system to another server. In this way, loading, processing, and reporting of the copied data do not impact the operational system's performance.

Offline Data Warehouse:

Data in the Datawarehouse is regularly updated from the Operational Database. The data in Datawarehouse is mapped and transformed to meet the Datawarehouse objectives.

Real time Data Warehouse:

In this stage, Data warehouses are updated whenever any transaction takes place in operational database. For example, Airline or railway booking system.

Integrated Data Warehouse:

In this stage, Data Warehouses are updated continuously when the operational system performs a transaction. The Datawarehouse then generates transactions which are passed back to the operational system.

Components of Data warehouse

Four components of Data Warehouses are:

Load manager: Load manager is also called the front component. It performs with all the operations associated with the extraction and load of data into the warehouse. These operations include transformations to prepare the data for entering into the Data warehouse.

Warehouse Manager: Warehouse manager performs operations associated with the management of the data in the warehouse. It performs operations like analysis of data to ensure consistency, creation of indexes and views, generation of denormalization and aggregations, transformation and merging of source data and archiving and baking-up data.

Query Manager: Query manager is also known as backend component. It performs all the operation operations related to the management of user queries. The operations of this Data warehouse components are direct queries to the appropriate tables for scheduling the execution of queries.

End-user access tools:

This is categorized into five different groups like 1. Data Reporting 2. Query Tools 3. Application development tools 4. EIS tools, 5. OLAP tools and data mining tools.

Who needs Data warehouse?

Data warehouse is needed for all types of users like:

- Decision makers who rely on mass amount of data
- Users who use customized, complex processes to obtain information from multiple data sources.
- It is also used by the people who want simple technology to access the data
- It also essential for those people who want a systematic approach for making decisions.
- If the user wants fast performance on a huge amount of data which is a necessity for reports, grids or charts, then Data warehouse proves useful.
- Data warehouse is a first step If you want to discover 'hidden patterns' of data-flows and groupings.

What Is a Data Warehouse Used For?

Here, are most common sectors where Data warehouse is used:

Airline:

In the Airline system, it is used for operation purpose like crew assignment, analyses of route profitability, frequent flyer program promotions, etc.

Banking:

It is widely used in the banking sector to manage the resources available on desk effectively. Few banks also used for the market research, performance analysis of the product and operations.

Healthcare:

Healthcare sector also used Data warehouse to strategize and predict outcomes, generate patient's treatment reports, share data with tie-in insurance companies, medical aid services, etc.

Public sector:

In the public sector, data warehouse is used for intelligence gathering. It helps government agencies to maintain and analyze tax records, health policy

records, for every individual.

Investment and Insurance sector:

In this sector, the warehouses are primarily used to analyze data patterns, customer trends, and to track market movements.

Retain chain:

In retail chains, Data warehouse is widely used for distribution and marketing. It also helps to track items, customer buying pattern, promotions and also used for determining pricing policy.

Telecommunication:

A data warehouse is used in this sector for product promotions, sales decisions and to make distribution decisions.

Hospitality Industry:

This Industry utilizes warehouse services to design as well as estimate their advertising and promotion campaigns where they want to target clients based on their feedback and travel patterns.

Steps to Implement Data Warehouse

The best way to address the business risk associated with a Datawarehouse implementation is to employ a three-prong strategy as below

1. **Enterprise strategy**: Here we identify technical including current architecture and tools. We also identify facts, dimensions, and attributes. Data mapping and transformation is also passed.
2. **Phased delivery**: Datawarehouse implementation should be phased based on subject areas. Related business entities like booking and billing should be first implemented and then integrated with each other.
3. **Iterative Prototyping**: Rather than a big bang approach to implementation, the Datawarehouse should be developed and tested iteratively.

Here, are key steps in Datawarehouse implementation along with its deliverables.

Step	Tasks	Deliverables
1	Need to define project scope	Scope Definition

2	Need to determine business needs	Logical Data Model
3	Define Operational Datastore requirements	Operational Data Store Model
4	Acquire or develop Extraction tools	Extract tools and Software
5	Define Data Warehouse Data requirements	Transition Data Model
6	Document missing data	To Do Project List
7	Maps Operational Data Store to Data Warehouse	D/W Data Integration Map
8	Develop Data Warehouse Database design	D/W Database Design
9	Extract Data from Operational Data Store	Integrated D/W Data Extracts
10	Load Data Warehouse	Initial Data Load
11	Maintain Data Warehouse	On-going Data Access and Subsequent Loads

Best practices to implement a Data Warehouse

- Decide a plan to test the consistency, accuracy, and integrity of the data.
- The data warehouse must be well integrated, well defined and time stamped.
- While designing Datawarehouse make sure you use right tool, stick to life cycle, take care about data conflicts and ready to learn you're your mistakes.
- Never replace operational systems and reports
- Don't spend too much time on extracting, cleaning and loading data.

- Ensure to involve all stakeholders including business personnel in Datawarehouse implementation process. Establish that Data warehousing is a joint/ team project. You don't want to create Data warehouse that is not useful to the end users.
- Prepare a training plan for the end users.

Why We Need Data Warehouse? Advantages & Disadvantages

Advantages of Data Warehouse:

- Data warehouse allows business users to quickly access critical data from some sources all in one place.
- Data warehouse provides consistent information on various cross-functional activities. It is also supporting ad-hoc reporting and query.
- Data Warehouse helps to integrate many sources of data to reduce stress on the production system.
- Data warehouse helps to reduce total turnaround time for analysis and reporting.
- Restructuring and Integration make it easier for the user to use for reporting and analysis.
- Data warehouse allows users to access critical data from the number of sources in a single place. Therefore, it saves user's time of retrieving data from multiple sources.
- Data warehouse stores a large amount of historical data. This helps users to analyze different time periods and trends to make future predictions.

Disadvantages of Data Warehouse:

- Not an ideal option for unstructured data.
- Creation and Implementation of Data Warehouse is surely time confusing affair.
- Data Warehouse can be outdated relatively quickly
- Difficult to make changes in data types and ranges, data source schema, indexes, and queries.
- The data warehouse may seem easy, but actually, it is too complex for the average users.
- Despite best efforts at project management, data warehousing project

scope will always increase.

- Sometime warehouse users will develop different business rules.
- Organisations need to spend lots of their resources for training and Implementation purpose.

What is Online Analytical Processing?

OLAP is a category of software that allows users to analyze information from multiple database systems at the same time. It is a technology that enables analysts to extract and view business data from different points of view. OLAP stands for Online Analytical Processing.

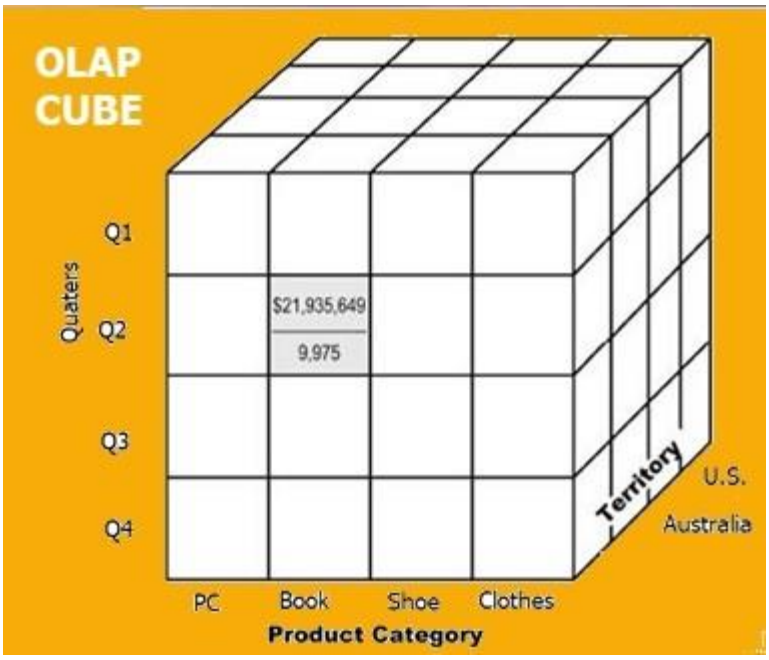
Analysts frequently need to group, aggregate and join data. These operations in relational databases are resource intensive. With OLAP data can be pre-calculated and pre-aggregated, making analysis faster.

OLAP databases are divided into one or more cubes. The cubes are designed in such a way that creating and viewing reports become easy.

You will learn-

- What is Online Analytical Processing?
- OLAP cube
- Basic analytical operations of OLAP
- Types of OLAP systems
- ROLAP
- MOLAP
- Hybrid OLAP
- Advantages of OLAP
- Disadvantages of OLAP

OLAP cube:



At the core of the OLAP, concept is an OLAP Cube. The OLAP cube is a data structure optimized for very quick data analysis.

The OLAP Cube consists of numeric facts called measures which are categorized by dimensions. OLAP Cube is also called the **hypercube**.

Usually, data operations and analysis are performed using the simple spreadsheet, where data values are arranged in row and column format. This is ideal for two-dimensional data. However, OLAP contains multidimensional data, with data usually obtained from a different and unrelated source. Using a spreadsheet is not an optimal option. The cube can store and analyze multidimensional data in a logical and orderly manner.

How does it work?

A Data warehouse would extract information from multiple data sources and formats like text files, excel sheet, multimedia files, etc.

The extracted data is cleaned and transformed. Data is loaded into an OLAP server (or OLAP cube) where information is pre-calculated in advance for further analysis.

Basic analytical operations of OLAP

Four types of analytical operations in OLAP are:

1. Roll-up
2. Drill-down
3. Slice and dice

4. Pivot (rotate)

1) Roll-up:

Roll-up is also known as "consolidation" or "aggregation." The Roll-up operation can be performed in 2 ways

1. Reducing dimensions
2. Climbing up concept hierarchy. Concept hierarchy is a system of grouping things based on their order or level.

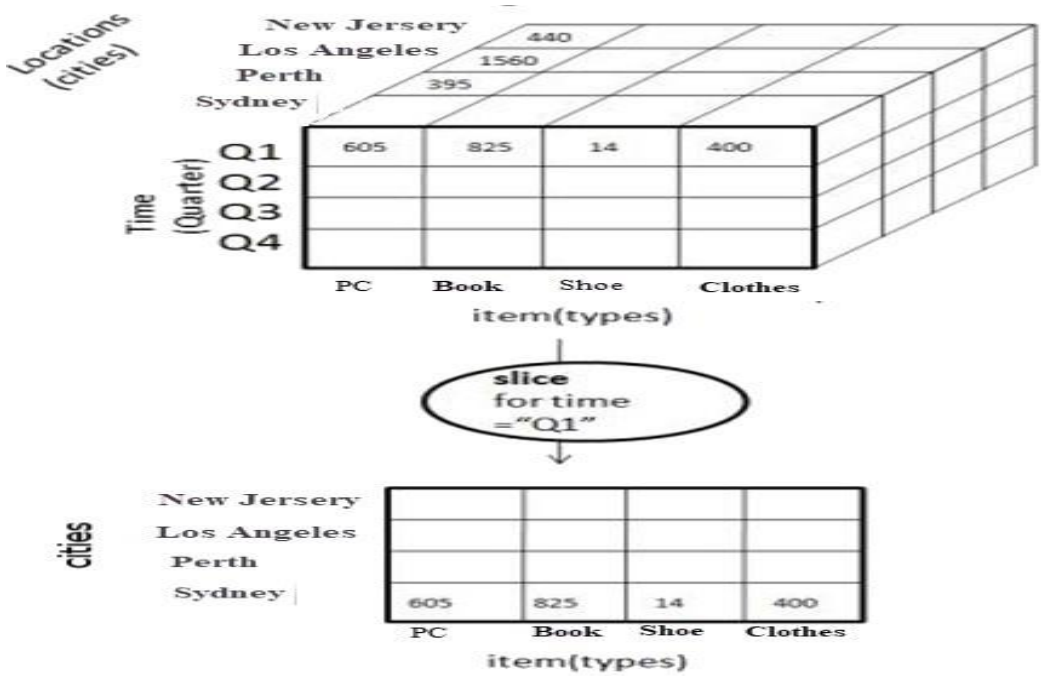
2) Drill-down

In drill-down data is fragmented into smaller parts. It is the opposite of the rollup process

3) Slice:

Here, one dimension is selected, and a new sub-cube is created.

Following diagram explain how slice operation performed:



- Dimension Time is Sliced with Q1 as the filter.
- A new cube is created altogether.

Dice:

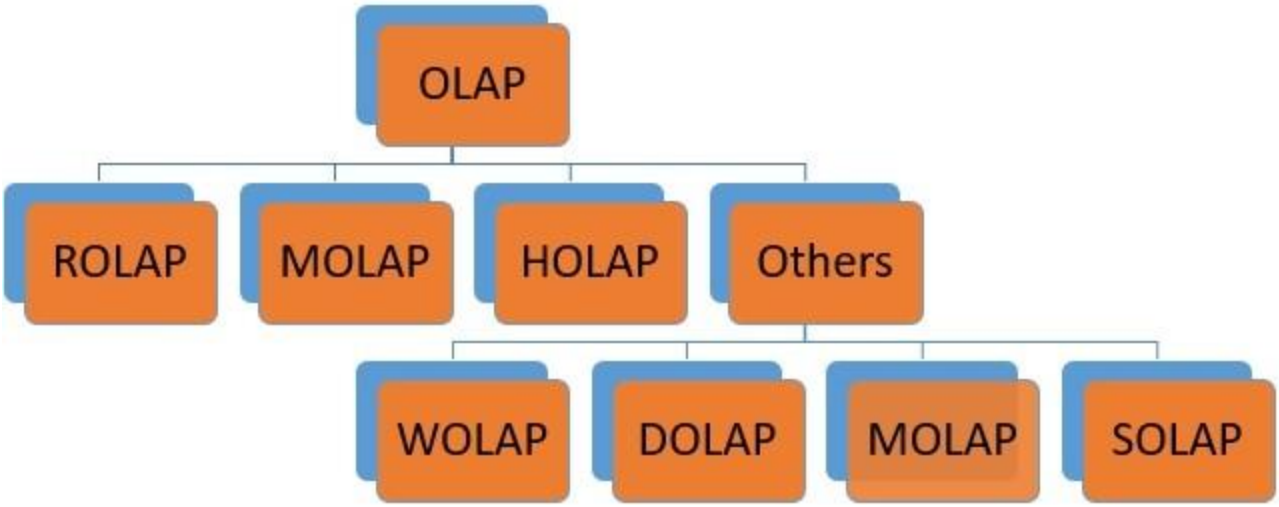
This operation is similar to a slice. The difference in dice is you select 2 or more dimensions that result in the creation of a sub-cube.

4) Pivot

In Pivot, you rotate the data axes to provide a substitute presentation of data.

Types of OLAP systems

OLAP Hierarchical Structure



Type of OLAP	Explanation
Relational OLAP(ROLAP):	ROLAP is an extended RDBMS along with multidi data mapping to perform the standard relational op
Multidimensional OLAP (MOLAP)	MOLAP Implementes operation in multidimensiona
Hybrid OnlineAnalytical Processing (HOLAP)	In HOLAP approach the aggregated totals are stor multidimensional database while the detailed data the relational database. This offers both data effici ROLAP model and the performance of the MOLAP
Desktop OLAP (DOLAP)	In Desktop OLAP, a user downloads a part of the d the database locally, or on their desktop and analy DOLAP is relatively cheaper to deploy as it offers v functionalities compares to other OLAP systems.

Web OLAP (WOLAP)	Web OLAP which is OLAP system accessible via t browser. WOLAP is a three-tiered architecture. It c three components: client, middleware, and a datab
Mobile OLAP:	Mobile OLAP helps users to access and analyze O using their mobile devices
Spatial OLAP :	SOLAP is created to facilitate management of both non-spatial data in a Geographic Information syste

ROLAP

ROLAP works with data that exist in a relational database. Facts and dimension tables are stored as relational tables. It also allows multidimensional analysis of data and is the fastest growing OLAP.

Advantages of ROLAP model:

- **High data efficiency.** It offers high data efficiency because query performance and access language are optimized particularly for the multidimensional data analysis.
- **Scalability.** This type of OLAP system offers scalability for managing large volumes of data, and even when the data is steadily increasing.

Drawbacks of ROLAP model:

- **Demand for higher resources:** ROLAP needs high utilization of manpower, software, and hardware resources.
- **Aggregately data limitations.** ROLAP tools use SQL for all calculation of aggregate data. However, there are no set limits to the for handling computations.
- **Slow query performance.** Query performance in this model is slow when compared with MOLAP

MOLAP

MOLAP uses array-based multidimensional storage engines to display multidimensional views of data. Basically, they use an OLAP cube.

Hybrid OLAP

Hybrid OLAP is a mixture of both ROLAP and MOLAP. It offers fast computation of MOLAP and higher scalability of ROLAP. HOLAP uses two

databases.

1. Aggregated or computed data is stored in a multidimensional OLAP cube
2. Detailed information is stored in a relational database.

Benefits of Hybrid OLAP:

- This kind of OLAP helps to economize the disk space, and it also remains compact which helps to avoid issues related to access speed and convenience.
- Hybrid HOLAP's uses cube technology which allows faster performance for all types of data.
- ROLAP are instantly updated and HOLAP users have access to this real-time instantly updated data. MOLAP brings cleaning and conversion of data thereby improving data relevance. This brings best of both worlds.

Drawbacks of Hybrid OLAP:

- Greater complexity level**:** The major drawback in HOLAP systems is that it supports both ROLAP and MOLAP tools and applications. Thus, it is very complicated.
- Potential overlaps**:** There are higher chances of overlapping especially into their functionalities.

Advantages of OLAP

- OLAP is a platform for all type of business includes planning, budgeting, reporting, and analysis.
- Information and calculations are consistent in an OLAP cube. This is a crucial benefit.
- Quickly create and analyze "What if" scenarios
- Easily search OLAP database for broad or specific terms.
- OLAP provides the building blocks for business modeling tools, Data mining tools, performance reporting tools.
- Allows users to do slice and dice cube data all by various dimensions, measures, and filters.
- It is good for analyzing time series.
- Finding some clusters and outliers is easy with OLAP.
- It is a powerful visualization online analytical process system which provides faster response times

Disadvantages of OLAP

- OLAP requires organizing data into a star or snowflake schema. These schemas are complicated to implement and administer
- You cannot have large number of dimensions in a single OLAP cube
- Transactional data cannot be accessed with OLAP system.
- Any modification in an OLAP cube needs a full update of the cube. This is a time-consuming process